

Transformative Entrepreneurs: Origins and Impacts on Economic Growth

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ABSTRACT

Business formation is often treated as a barometer of dynamism, yet startups differ sharply in their capacity to innovate and grow. Using Danish administrative data linking founders to family background, cognitive ability, education, firms, and workers, we show that aggregate entrepreneurial entry declines with cognitive ability. The minority who hire R&D workers (i.e., transformative entrepreneurs) are positively selected on ability and education and build larger, faster-growing firms. We develop and estimate an endogenous-growth model in which occupational choice jointly determines the supply of R&D talent and the demand for it from heterogeneous founders. In the calibrated economy, at a common fiscal cost, support for transformative entrants raises long-run growth more than twice as much as support for incumbent R&D, whereas uniform entry support has little effect. Firm composition, not entry alone, matters for growth because innovative entry and the supply of R&D talent reinforce one another.

Keywords: Entrepreneurship, Firm Dynamics, Innovation, Endogenous Growth, Occupational Sorting.

JEL Classification: O31, O38, O47, J24.

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1 Introduction

Joe’s Electronics, a repair shop on the corner, and Hewlett-Packard (HP) both began as entrepreneurial ventures. As demand grows, Joe’s hires another technician to repair more equipment, but is likely to remain a small local business, operating largely with existing technologies. HP followed a different path: it hired engineers, invested in R&D, developed and patented new technologies, and grew into a global technology company. Business formation statistics count each as one startup. Economic growth does not.

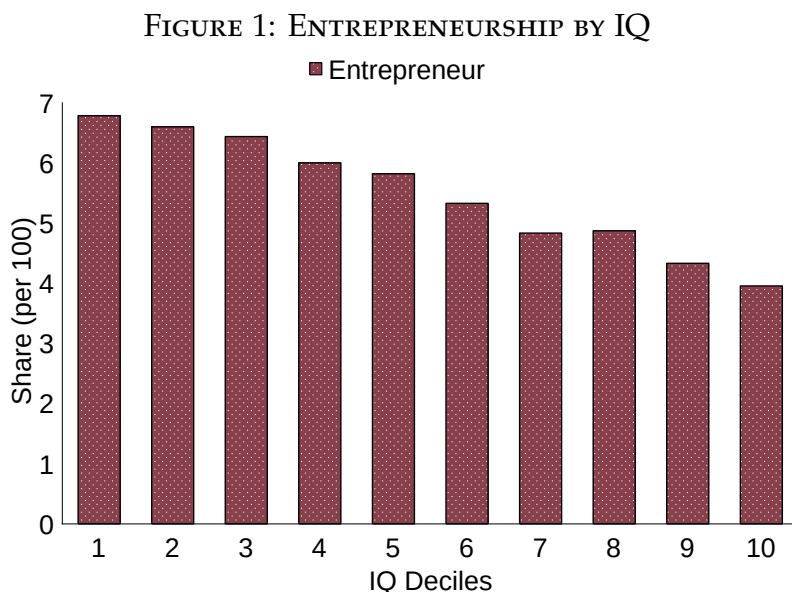
This distinction is obscured by aggregate measures of entrepreneurship. Business formation is widely used as a barometer of economic dynamism, and canonical growth theories place new firms at the center of innovation and creative destruction (e.g., Romer, 1990; Aghion and Howitt, 1992). Yet more entrepreneurship need not translate into more innovation or growth. U.S. business applications rose from 3.5 million in 2019 to nearly 5.4 million in 2021, a surge celebrated as a “historic small business boom” (The White House, 2022). But the employment footprint of the smallest firms has barely changed: employment at businesses with 1–9 workers is less than 2% above its 2019 level, compared with an almost 5% increase in total nonfarm employment (Intuit QuickBooks, 2026; U.S. Bureau of Labor Statistics, 2026).¹ Thus, even a historic surge in business formation need not generate a commensurate expansion in small-business employment. For economic growth, the composition of entrepreneurship may matter far more than its quantity.

The relevant question, therefore, is not only how many entrepreneurs an economy creates, but also what types of entrepreneurs it creates. What distinguishes the small minority who build innovation-intensive firms? Where do they come from, and how does their demand for innovative talent interact with its supply? We answer these questions theoretically, empirically, and quantitatively, using Danish administrative microdata linking individuals’ family background, education, and cognitive ability to the firms they found.

Who becomes an entrepreneur in the first place? Economists’ default answer is selection on ability: people sort into the occupations that fit their talents, so demanding occupations attract the most able. Even though we like to watch and listen, no amount of effort would make any of the authors of this paper NBA players or concert pianists, and the people who fill those jobs are drawn overwhelmingly from the far right tail of the relevant ability distribution. Cognitive ability strongly predicts entry into the economy’s other innovative occupations (science, invention, research), but for entrepreneurship, the

¹Businesses with 1–9 workers employed 12.55 million workers in 2019 and 12.79 million in July 2026. Over roughly the same period, total nonfarm employment increased from 151.79 million to 158.86 million.

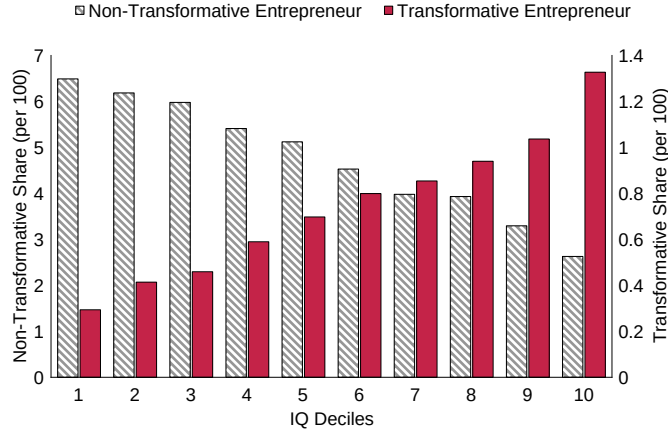
relationship is not merely weaker: it is inverted, as shown in Figure 1. The probability of becoming an entrepreneur, defined as the primary founder of a firm with at least one employee, declines with cognitive ability, a pattern that is strong, robust across specifications, and to our knowledge new to the literature. The economy's entrepreneurs are not drawn from the tail of the talent distribution; they are drawn away from it.



Notes: Share of individuals in each IQ decile who become entrepreneurs, defined as primary founders of firms with at least one employee, incorporated or unincorporated. Source: Statistics Denmark (DST).

The entrepreneurs who transform industries, however, are a small minority of those who start businesses, and overall entry can fall with ability even as that minority rises with it. We therefore separate entrepreneurs into two types by their *intent* to innovate, revealed by whether they hire workers with the capacity for invention: R&D workers, defined as those in occupations with high patenting rates. Only 17% of entrepreneurs hire them; we call these entrepreneurs *transformative*, and the rest non-transformative. Figure 2 plots the two types separately across the ability distribution, and shows that they sort in opposite directions: non-transformative entry falls steeply with IQ while transformative entry rises. Among entrepreneurs in the bottom IQ decile, only 4% are transformative, compared to 33% in the top decile. The decline in overall entry in Figure 1 comes entirely from the non-transformative margin: pooling the two types hides the minority that innovates.

FIGURE 2: TRANSFORMATIVE AND NON-TRANSFORMATIVE ENTREPRENEURS BY IQ



Notes: Share of individuals in each IQ decile who become non-transformative entrepreneurs (left axis) and transformative entrepreneurs (right axis). Transformative entrepreneurs hire at least one R&D worker; non-transformative entrepreneurs hire none. *Source*: DST.

We then examine the characteristics of the transformative entrepreneurs and document three main empirical facts that motivate the model:

Fact 1. The likelihood of becoming an entrepreneur rises with parental entrepreneurship more than with any other characteristic, while it *declines* with IQ.

Fact 2. IQ and education strongly predict becoming a transformative entrepreneur.

Fact 3. The firms of transformative entrepreneurs grow faster and create more jobs than those of non-transformative entrepreneurs.

Children of entrepreneurs are substantially more likely to become entrepreneurs themselves than children from other backgrounds. Conditional on covariates, a college degree raises the odds of being a transformative entrepreneur by 55% and top-quartile IQ raises them by 110%, while the negative association between IQ and non-transformative entry survives the same controls. The distinction matters greatly for job creation: by their tenth year, transformative firms grow to more than five times their founding size, while non-transformative firms merely double in size.

Guided by these facts, we develop an endogenous growth model in which the supply of and demand for innovative talent are jointly determined by occupational choice. Individuals differ in ability, family income, and exposure to entrepreneurship through their parents. Higher education is needed to an R&D career and must be paid out of family resources, so financial frictions affect talented children of poorer families. Entrepreneurial opportunities arrive over the life cycle, and ability together with education determines whether the opportunity is transformative. Only transformative entrepreneurs hire R&D

workers and innovate, with decreasing returns to innovation at the firm level (Akcigit and Kerr, 2018).

The model's central mechanism is the market for innovative talent, and the two sides of this market determine each other: an R&D career pays only when transformative entrepreneurs are hiring (Michelacci, 2003), and transformative entrepreneurs can innovate only when there are R&D workers to hire. The two sides are linked over the life cycle as well: R&D workers are the individuals most likely to become transformative entrepreneurs, so today's supply of innovative talent also creates tomorrow's demand for it. Financial frictions in access to education can transmit to both sides at once, because the same constrained families lose future R&D workers and future transformative entrepreneurs together. A policy aimed at one side of this market therefore moves the other.

We estimate the model by Simulated Method of Moments, using the Danish data on firms and individual backgrounds. We match occupation shares by IQ, the selection of transformative entrepreneurs on IQ and education, and the life-cycle growth of the two types of firms. Out of sample, the model reproduces transition rates from prior occupations into the two entrepreneur types and the enrollment response to the 1988 Danish stipend reform studied by Nielsen et al. (2010). We then use the estimated model to compare policies on both sides of the market for innovative talent at a common fiscal cost: demand-side policies that subsidize firms' R&D or entry, and supply-side policies that subsidize education. Three conclusions follow.

First, on the demand side, the type of firm targeted by a subsidy matters for its effectiveness. At a budget of 0.025% of GDP, a subsidy targeted to transformative entrants raises the growth rate by 5.06% of its baseline level, more than double the 2.39% from an R&D subsidy to incumbents. A uniform entry subsidy delivers 0.10%: it mostly funds non-transformative entrants rather than the firms that innovate. In practice, policymakers rely on noisy signals to identify an entrant's type, so we also quantify how much of the gain survives imperfect classification.

Second, targeting matters on the supply side as well. At the same budget, a full scholarship aimed at low-income, high-ability students raises growth by 8.69%, against 6.72% for scholarships allocated at random; an income screen alone captures most of the difference, while an ability screen alone adds almost none of it. The joint screen does best because it reaches students who are both constrained and high-return: financial frictions exclude low-income students from education, while most high-ability students from richer families enroll anyway.

Third, the two sides of the market are inseparable in general equilibrium. Every policy we study needs both sides of the talent market to have its full effect. Demand-side

policies depend on education. With the supply of educated workers held at its baseline, the R&D subsidy raises growth by 0.03% instead of 2.39%, and the transformative entry subsidy by 1.18% instead of 5.06%. Supply-side policies depend on firm entry in the same way. Holding entry at its baseline removes over 40% of the growth gain from each of them, and the policies that achieve the most on the supply side alone are also the ones that gain the most from entry. On both sides of the market, the most effective policy is the one that moves both.

Taken together, the results reframe what innovation policy can accomplish. The quantity of business formation is close to irrelevant for growth; the allocation of talent across the two kinds of entrepreneurship, and between entrepreneurship and R&D, is what matters. And because that allocation is set on both sides of the market for innovative talent at once, no policy on either side has its full effect unless the other side can respond. The rest of the paper proceeds as follows. The remainder of this section reviews the literature. Section 2 introduces the theoretical model. Section 3 presents the empirical evidence. Section 4 connects the theory and data by estimating the model. Section 5 evaluates policies on the demand and supply sides of the market for innovative talent. Section 6 concludes.

Literature Review. Our paper connects several strands of work. First, in Schumpeterian growth models, firm dynamics and creative destruction link innovation to aggregate outcomes (Schumpeter, 1911; Aghion and Howitt, 1992; Klette and Kortum, 2004; Lentz and Mortensen, 2008; Akcigit and Kerr, 2018; Garcia-Macia et al., 2019), with recent contributions documenting the secular decline in business dynamism (Akcigit and Ates, 2021; Aghion et al., 2023; Andrews et al., 2015; De Ridder, 2024; Olmstead-Rumsey, 2022).² These models typically abstract from the nature of the individual entrepreneurs who found and lead these firms, leaving firm dynamics disconnected from the characteristics and choices of their owners. We build on this literature by embedding entrepreneurial heterogeneity within a general-equilibrium framework in which both the supply of and the demand for innovative talent are endogenously determined by occupational choice, with feedback between the two. This formulation opens a new dimension for policy, allowing us to analyze how interventions that target the composition of the entrepreneur population can reshape firm dynamics and growth. It also lets us ask how policies on the demand and supply sides of this market depend on each other.

Second, research in organizational economics studies how managerial talent, knowledge hierarchies, and firm structure shape productivity and scale (Lucas, 1978; Gari-

²For additional readings on declining dynamism, see Decker et al. (2020); Akcigit and Ates (2023); Karahan et al. (2024); Hopenhayn et al. (2022), among others.

cano, 2000; Garicano and Rossi-Hansberg, 2006; Bloom et al., 2021). A complementary literature emphasizes how occupational choice and the allocation of talent affect development and growth (Baumol, 1990; Murphy et al., 1991; Gennaioli et al., 2013; Akcigit et al., 2025; Prato, 2025). In this vein, Mestieri et al. (2017) highlight the role of early human-capital accumulation in entrepreneurial sorting, Queiró (2022) shows that more educated entrepreneurs build larger and faster-growing firms, primarily through technology adoption, Morazzoni and Sy (2022) quantify the losses from gender-based misallocation in entrepreneurship, and Hsieh et al. (2019) document the adverse impact of race- and gender-based discrimination on aggregate growth. Relative to these papers, we integrate background characteristics (ability, education, parental environment) into a quantitative macro model in which transformative entrepreneurs hire R&D workers, thereby tying organizational demand for scientific talent to aggregate growth and policy (Atkeson and Burstein, 2019; Liu and Ma, 2021).

This paper studies a distinct margin from Akcigit et al. (2025). Whereas that paper asks how education and innovation policy shape the supply of innovative talent (i.e., who acquires advanced skills and becomes a researcher or inventor) we ask who founds the firms that organize and employ that talent. Using linked Danish founder–worker data, we show that entrepreneurship in the aggregate is negatively selected on cognitive ability, whereas entry into innovation-intensive firms is positively selected on ability and education. These founders build systematically larger and faster-growing firms and generate demand for R&D workers. We embed this organizational margin in a growth model with heterogeneous founders, firm entry and exit, and endogenous demand for innovative talent. This leads to a distinct policy question: when is it more effective to support incumbent R&D, broad entry, or high-potential innovation-intensive entry? Education matters in this framework because it shapes the pool of people who can become both R&D workers and innovation-intensive founders, thereby influencing the formation and growth of the firms that deploy innovative talent.

Third, a large empirical literature studies who becomes an entrepreneur and what they do. Work on cognitive ability and entry finds ability to be a weak or ambiguous predictor when the self-employed are treated as a single group, with the ambiguity resolving once they are disaggregated: Levine and Rubinstein (2017) show that those who incorporate score highly on ability and learning aptitude while the unincorporated do not, and Hartog et al. (2010) find that components of ability carry different returns in self-employment than in wage work. We document something stronger than ambiguity: pooled entry is *negatively* selected on ability, across the full ability distribution and measured with a test taken before any occupational choice is made. The underlying distinction between high-impact and low-impact entrepreneurship is itself well established

(Schoar, 2010; Hurst and Pugsley, 2011), with prior work proxying growth orientation by legal form (Levine and Rubinstein, 2017), venture capital (Puri and Zarutskie, 2012), predictive indicators (Guzman and Stern, 2020), or realized employment growth (Decker et al., 2014). We contribute on both fronts. We classify entrepreneurs by innovation intent revealed through an input choice: whether the founder hires an R&D worker. This measure captures commitment to knowledge production directly, rather than inferring it from legal status or realized growth, and it correlates with prior R&D employment, operation in a high-tech industry, and incorporation. Applying that classification, we find that transformative entrepreneurs indeed have higher IQ and more education, so the negative pooled gradient and this positive one are two sides of one fact rather than competing findings. Our measure also connects to evidence on the backgrounds and impacts of inventors (Akcigit et al., 2017; Bell et al., 2018) and allows us to study founders and the R&D workers they hire as two sides of one market.

Our empirical analysis is informed by work on selection into entrepreneurship. Family and cultural background, human capital, and resources predict entry (Evans and Jovanovic, 1989; Blanchflower and Oswald, 1998; Doepke and Zilibotti, 2014; Levine and Rubinstein, 2017). Organizational behavior and sociology emphasize parental entrepreneurship and early exposure (Sørensen, 2007; Hvide and Oyer, 2018; Sorenson et al., 2021); twin, sibling, and adoption studies point to both genetic and environmental channels (Lindquist et al., 2015; Vladasels et al., 2021; Vladasels, 2023). Local entrepreneurial density fosters managerial practices and entry (Guiso et al., 2021). Leadership potential matters (Hartog et al., 2010), and non-pecuniary returns are not central, consistent with selection on comparative advantage and growth ambitions (Bhandari et al., 2026).

Finally, our mechanism relates to team formation and the demand for innovative labor. Entrepreneurs' generalist skills, networks, and financing shape which ideas are commercialized and which teams are formed (Lazear, 2004; Hochberg et al., 2007, 2010; Kaplan et al., 2012). When entrepreneurial demand is weak, returns to an R&D career are low (Michelacci, 2003). Lindner et al. (2022) find that innovating firms raise both their skill ratio and the skill premium they pay, suggesting the demand side plays a central role in innovation. The rise of inventor teams is another channel through which entrepreneurship can play a major role (Wuchty et al., 2007; Pearce, 2020; Freund, 2023). We bring these insights into a unified macro framework in which transformative entrepreneurs create the demand for scientists and engineers, linking individual backgrounds, occupational choice, team composition, and firm dynamics to aggregate innovation and growth. In doing so, we provide new evidence on selection into the two types of entrepreneurship and quantify how policies on the supply side of the market

for innovative talent, through education, and on its demand side, through support to growth-oriented startups and their R&D, depend on each other.

2 Model

This section develops a life-cycle model of career choice, entrepreneurship, and firm dynamics embedded in an endogenous growth framework, guided by the empirical patterns that we document later in Section 3. In our model, individuals differ in ability, family income, and exposure to parental entrepreneurship. These differences affect schooling decisions and ultimately shape occupational choices, the evolution of firms, aggregate innovation, and economic growth. The model is formulated in continuous time and designed to capture the interplay between talent allocation across occupations and firm dynamics along a Balanced Growth Path (BGP) equilibrium. In our exposition of the model, we normalize all the growing variables by the final good in the economy to keep the stationary equilibrium values constant, and suppress time subscripts whenever this causes no confusion. Figure 3 illustrates the key elements of the model, from initial career choice between R&D and production worker, to entrepreneurial choice, to the dynamics of firms and innovation.

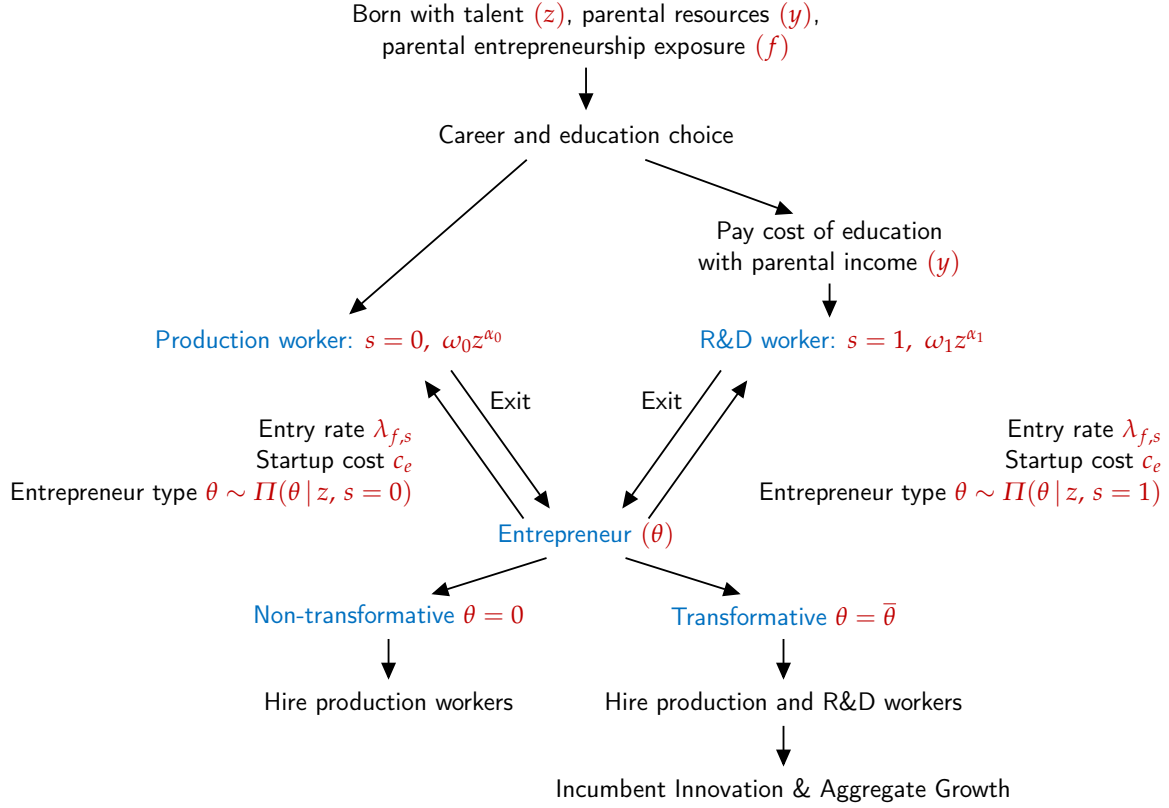
2.1 Environment and Agent Heterogeneity

We consider an economy populated by agents that are heterogeneous in three dimensions: (i) worker ability (z), (ii) parental resources (y), and (iii) exposure to entrepreneurship through parents (f). In every period, a measure m of new agents enters the economy, with characteristics drawn from the joint distribution $\Omega(z, y, f)$. Agents die at an arrival rate ψ and are immediately replaced by new agents with the same characteristics such that the distribution of characteristics in the economy remains unchanged.

2.2 Educational Choice and Occupational Sorting

When individuals are born, they initially decide on an occupation to begin their working career, choosing either to become a production worker or an R&D worker. An R&D career requires the acquisition of R&D-qualifying schooling at cost c_s ; by contrast, produc-

FIGURE 3: MODEL MAP



tion workers do not acquire such schooling.³ We interpret this schooling cost broadly: not necessarily as tuition fees, but rather as living expenses, books and supplies, the opportunity cost of forgone income during schooling, and other related expenditures. Importantly, we assume that this cost must be paid in advance through parental resources, reflecting a form of financial friction.⁴ Therefore, individuals with $y < c_s$ cannot afford schooling, and they become production workers. In addition, individuals have prefer-

³For expositional simplicity, the model treats “schooling” as acquisition of R&D-qualifying human capital that deterministically yields R&D employment. Accordingly, throughout the paper, “schooling” should be interpreted as the relevant post-secondary or technical education enabling R&D work. Consistent with this interpretation, the estimation in Section 4 matches the model’s educated group to individuals in R&D occupations. In the empirical analysis of Section 3, we instead proxy schooling with a college indicator: although college education implies a much higher probability of R&D employment and of transformative entrepreneurship, the mapping is imperfect, as some college graduates work in production. The framework could be generalized to a probabilistic assignment in which schooling increases the likelihood of R&D employment, thereby allowing schooled individuals to remain in production, but we retain the present specification for parsimony and tractability.

⁴The literature has documented the presence of financial frictions and the influence of parental income on access to college education in Denmark, despite the low cost of education compared to other countries; see Nielsen et al. (2010) and Akcigit et al. (2025). We will use the evidence from Nielsen et al. (2010) to calibrate the magnitude of these financial frictions, which play a central role in the allocation of talent within our framework.

ences over schooling characterized by an idiosyncratic preference shock ϵ .

We denote the choice of schooling with the indicator variable $s \in \{0, 1\}$, which takes the value 1 for an individual who acquires schooling, equivalent to becoming an R&D worker in the model.⁵ Given this structure, the occupation choice problem of an individual with state (z, y, f) who can afford schooling can be expressed as:

$$\max \{W_0(z, y, f), W_1(z, y, f) - c_s + \epsilon\}. \quad (1)$$

Here, $W_s(z, y, f)$, $s \in \{0, 1\}$, which will be derived in the next subsection, denotes the lifetime values associated with a career as a production worker ($s = 0$) and an R&D worker ($s = 1$), excluding schooling cost and preference shock. We assume the preference shock follows a logistic distribution with mean χ and scale parameter ς .⁶ Given this distributional assumption, the above decision problem implies that the probability of acquiring schooling and becoming an R&D worker is given by

$$\mathbb{P}(s = 1|z, y, f) = \mathbb{I}(c_s < y) \times \frac{1}{1 + \exp\left(\frac{W_0(z, y, f)}{\varsigma} - \frac{W_1(z, y, f) - c_s + \chi}{\varsigma}\right)}. \quad (2)$$

This equation highlights two key channels through which initial career selection operates: (i) the direct role of financial resources in restricting education access and (ii) the economic tradeoffs between present schooling costs and future career returns. The first component, $\mathbb{I}(c_s < y)$, is an indicator function that equals 1 if parental resources exceed the cost of schooling and 0 otherwise. This represents the financial constraint that individuals face: education is only possible when family resources can cover its costs. This formulation directly captures the credit market imperfections that prevent talented individuals from low-resource backgrounds from obtaining education.⁷ The second component, a standard logit function derived from the distribution of preference shocks, reflects the tradeoff between the lifetime values of being a production worker $W_0(z, y, f)$ and an R&D worker $W_1(z, y, f)$, adjusted for schooling costs c_s and average preference for schooling χ . A higher ς increases the role of idiosyncratic preferences, making the decision more dispersed and less deterministic.

⁵Accordingly, in our model, the choice of schooling is effectively equivalent to choosing between a career as a production worker ($s = 0$) or an R&D worker ($s = 1$).

⁶In this setting, a negative value for χ can be interpreted as disutility from schooling.

⁷We make two simplifying assumptions here, both for tractability: the friction takes the form of a hard borrowing constraint, and family resources can be used only to finance schooling rather than spent freely. A richer formulation would relax both, letting parental resources be allocated between consumption and schooling and replacing the sharp cutoff with imperfect borrowing, for instance by borrowing against future earnings up to a limit or at an interest rate that rises with the amount borrowed (Keane and Wolpin, 2001; Chatterjee et al., 2007; Lochner and Monge-Naranjo, 2011). This would preserve the essential feature of our friction, that access to education depends on family resources and not on ability alone.

After making the initial career choice, agents enter the competitive labor market where they start earning wage income $\omega(z, s)$ based on their ability z and occupation of production worker or R&D worker, corresponding to $s = 0$ and $s = 1$ respectively. We allow for heterogeneity in how ability affects worker productivity across occupations by assuming that a worker with ability z supplies z^{α_s} efficiency units in the labor market, where α_s governs the rate at which productivity increases with ability for a given occupation of production worker ($s = 0$) or R&D worker ($s = 1$). Consequently, the wage income per period is given by

$$\omega(z, s) \equiv \omega_s z^{\alpha_s}$$

where ω_s denotes the wage *per efficiency unit* for a production worker ($s = 0$) and an R&D worker ($s = 1$), which will be determined in equilibrium by the set of entrepreneurs and their firms, which we turn to in the following section.

2.3 Entrepreneurial Opportunities and Startup Decisions

In addition to earning wage income, agents receive random opportunities to become entrepreneurs and earn profits from running a firm over the course of their life cycle. This opportunity arrives at rate $\lambda_{f,s}$, which depends on whether the agent comes from an entrepreneurial family ($f = 1$) or not ($f = 0$), and on their schooling $s \in \{0, 1\}$. When the entrepreneurship opportunity arises, the agent draws an entrepreneur type θ from a distribution $\Pi(\theta|z, s)$ that depends on the agent's ability and education, linking worker background to entrepreneurial type. The drawn type θ determines the entrepreneur's efficiency in utilizing R&D workers and their contribution to the firm's innovation process, thereby shaping entrepreneurial returns and firm performance, which will be explained further in the next subsection. We assume θ can take two possible values, $\theta \in \{0, \bar{\theta}\}$. Entrepreneurs of type $\bar{\theta}$ are *transformative entrepreneurs*: they hire both production workers and R&D workers, and the latter enable them to engage in innovation activity in an attempt to expand their firms. Entrepreneurs of type $\theta = 0$ are only involved in production and do not innovate, but they may still grow their firms exogenously.

An individual's decision to transition from worker to entrepreneur involves comparing the benefits of starting a firm, net of a stochastic entry cost c_e , with the continuation value of remaining a worker.⁸ If they decide to become an entrepreneur, they leave the labor force and start a firm. Thus, an agent can occupy one of three roles at any given

⁸Here, the cost of starting a new firm is orthogonal to family background, which is consistent with the empirical evidence from the literature (Robb and Robinson, 2012).

time: (i) production worker, (ii) R&D worker, and (iii) entrepreneur.⁹ Given this setting, the lifetime value of the workers solves the following equation:

$$\begin{aligned} \rho W_s(z, y, f) = & \omega(z, s) \\ & + \lambda_{f,s} [\mathbb{E}_{c_e, \theta} (\max \{V_e(\theta) - c_e, W_s(z, y, f)\}) - W_s(z, y, f)] \\ & + \psi [0 - W_s(z, y, f)] \end{aligned} \quad (3)$$

where ρ is the discount rate.¹⁰ The left-hand side, $\rho W_s(z, y, f)$, represents the required return on the worker's lifetime value at the discount rate ρ , which equals the sum of three distinct components. First, the worker earns a per-period wage, $\omega(z, s)$, based on their ability and education. Second, workers have the option to transition into entrepreneurship when opportunities arrive at the rate $\lambda_{f,s}$; the term inside brackets captures the expected additional benefit from potentially becoming an entrepreneur, considering both the entrepreneurial payoff $V_e(\theta)$ (defined explicitly in the next section) net of entry costs c_e , and the continuation value of remaining a worker $W_s(z, y, f)$. This expectation is taken over the distributions of the entry cost and entrepreneurial type θ . Finally, the last term accounts for mortality: workers exit the economy at rate ψ , at which point their lifetime value drops to zero and a new worker enters. We next turn to describe what entrepreneurs do in detail and the resulting firm dynamics.

2.4 Entrepreneurship and Firm Dynamics

The firm side of the model builds on the framework of Klette and Kortum (2004) to capture the dynamics of innovation and firm growth. In our setting, each firm is owned and managed by an entrepreneur and is conceptualized as a portfolio of product lines that generate profits via employing production workers.¹¹ Entrepreneurs start their firms with a single product line, reflecting the minimal initial scale of new firms. Firms invest in innovation, using R&D workers as input, to expand their portfolios by improving product quality. While innovation grows firm value, this growth is offset by competition

⁹We abstract from financial frictions in the occupational choice between entrepreneurship and employment. Danish evidence indicates that access to finance is not a binding constraint for most potential entrepreneurs (Andersen and Nielsen, 2012; Jensen et al., 2022). Moreover, as we show later in the paper, family income is not a significant predictor of entrepreneurial entry in our data. Thus, for this paper, we focus on the less-studied education friction.

¹⁰As discussed in Appendix A.1, agents belong to representative households that pool income and insure idiosyncratic career risks; career decisions therefore maximize expected income rather than reflecting individual risk preferences. Due to the normalization of value functions by the final good, the value function is discounted with ρ rather than interest rate r ; see Appendix A.4 for details.

¹¹These product lines are used as inputs to produce the final good in the economy. We provide further details in Appendix A.2.

from incumbents and entrants that displace product lines. As a result, firm value evolves endogenously with the capture of new product lines and the loss of existing ones.

The firm's dynamic decision is characterized by its choice of innovation effort, which determines the arrival rate of new product lines. A firm with n product lines and entrepreneur type θ produces innovations at rate

$$X(n, \theta) = \theta l_{rd}^\sigma n^\eta, \quad (4)$$

where l_{rd} denotes the efficiency units of R&D labor hired by the firm, σ captures the elasticity of innovation with respect to R&D labor, and η governs returns to scale in the number of product lines.¹² The parameter θ reflects entrepreneurial innovation capability: transformative entrepreneurs have $\theta = \bar{\theta} > 0$ and actively invest in R&D, while non-transformative entrepreneurs have $\theta = 0$ and therefore do not attempt to innovate and do not hire any R&D workers. Given this technology, the cost of producing innovation effort X is given by the cost of the R&D workers it requires:

$$C(X; n, \theta) = \omega_1 \left(\frac{X}{\theta n^\eta} \right)^{\frac{1}{\sigma}}$$

where ω_1 is the wage per efficiency unit of R&D labor.

We allow for decreasing returns to scale in innovation, $\sigma + \eta \leq 1$, motivated by the evidence in Akcigit and Kerr (2018). This assumption plays an important role in the policy analysis of Section 5. In the polar case of constant returns ($\sigma + \eta = 1$), the innovation produced from a given amount of R&D labor is independent of how product lines are divided across firms, so concentration is costless; and because each entrepreneur is one fewer person available for production or R&D, the growth-maximizing arrangement would place all product lines under a single entrepreneur and everyone else in the labor force. Decreasing returns overturn this. Because each firm faces diminishing returns to its own innovation, a given amount of R&D produces more when it is spread across more firms than concentrated in a few large ones, so there is a benefit to having more innovating firms. This benefit must be weighed against the production and R&D labor that those entrepreneurs would otherwise supply, and the growth-maximizing allocation of talent between running firms and working in them balances the two. The demand-side policies of Section 5 act on this tradeoff in different ways: an R&D subsidy raises innovation at existing firms, while entry subsidies encourage the entry of new ones.

When innovation is successful, the firm improves the quality of a random product

¹²The inclusion of n in the innovation function reflects that firms with more product lines have a larger base of knowledge and resources, enhancing their capacity to generate innovations.

in the economy by a factor γ and captures it from another firm, thereby increasing its portfolio by one product. Symmetrically, each product line the firm owns is at risk of being displaced by other firms and entrants at the rate τ . We refer to τ as the *business reallocation rate*, which is endogenously determined in equilibrium and taken as given by the firm.

The value of a firm with n product lines and type θ is denoted by $V(n, \theta)$ and satisfies the following Bellman equation:

$$\begin{aligned} \rho V(n, \theta) = & \max_X \left\{ \pi n - \omega_1 \left(\frac{X}{\theta n^\eta} \right)^{\frac{1}{\sigma}} \right. \\ & + (X + \nu n) [V(n+1, \theta) - V(n, \theta)] \} \\ & + \tau n [V(n-1, \theta) - V(n, \theta)] + \psi [0 - V(n, \theta)]. \end{aligned} \quad (5)$$

The first line captures the static returns, given by πn , or the total flow profits generated by the n product lines¹³, minus the R&D investment. The second line captures the expansionary drivers of the firm, through endogenous innovation at rate X or exogenous factors at rate ν . Both terms lead to a change in the firm value as the firm expands its portfolio (increasing the number of product lines from n to $n+1$). The next term reflects the loss in value due to competition by other incumbents and new entrepreneurs, where each product line faces the risk of being displaced at the rate τ . Finally, the last term accounts for the exit due to the death of the entrepreneur, in which case the value of the firm goes to 0. Note that since firms enter with a single product line, the value of choosing entrepreneurship introduced in the previous section satisfies $V_e(\theta) = V(1, \theta)$.

Firm exit occurs either exogenously through the death of the entrepreneur or endogenously when all product lines of the firm are lost due to competition from other firms. In the latter case, the entrepreneur shuts down the firm and returns to the labor market in their previous occupation.¹⁴ Together, the entry and exit processes generate a dynamic equilibrium in which the composition and size of firms continuously evolve, thereby driving aggregate growth and the reallocation of resources across the economy.

2.5 Labor Market Clearing and Stationary Equilibrium Distributions

One of the main novel elements of this framework is connecting the origins of the supply of and demand for innovative talent. This section discusses the labor market clearing that

¹³Details on the product market and the resulting profits πn are provided in Appendix A.2.

¹⁴We assume $V(0, \theta) = 0$ to simplify the state space of the entrepreneur's value function. The stationary distributions in Appendix A.5 still account for these exits, with exiting entrepreneurs returning to the worker pool in their original occupation.

enables this general equilibrium treatment. On the supply side of the labor market, recall that each individual with ability z who works as a production or R&D worker supplies z^{α_s} efficiency units of labor, and the total number of such workers is determined by agents' educational choices early in life and their entrepreneurial decisions over the life cycle. On the demand side, firms demand $\frac{1}{\omega_0 \gamma}$ efficiency units of production worker per product line they own, which is derived in Appendix A.2. Demand for R&D workers, on the other hand, is determined by firms' innovation effort choice X implied by their dynamic problem (5).

Let $\Phi_s(z, y, f)$ denote the density of individual workers with talent z , parental income y , family entrepreneurship background f , and schooling $s \in \{0, 1\}$. Let $\Psi(n, \theta)$ denote the joint distribution of active firms by number of product lines n and entrepreneurial type θ . Labor market clearing then requires that demand and supply equalize with

$$\sum_{\theta} \sum_n \frac{1}{\omega_0 \gamma} n \Psi(n, \theta) = \sum_f \int_z \int_y \Phi_0(z, y, f) z^{\alpha_0} dz dy \quad (6)$$

for production workers, and

$$\sum_{\theta} \sum_n \left(\frac{X(n, \theta)}{\theta n^{\eta}} \right)^{\frac{1}{\sigma}} \Psi(n, \theta) = \sum_f \int_z \int_y \Phi_1(z, y, f) z^{\alpha_1} dz dy \quad (7)$$

for R&D workers. To solve for these labor market clearing conditions, we need to characterize both the worker and firm distributions, $\Phi_s(z, y, f)$ and $\Psi(n, \theta)$. We provide the derivation of these distributions in Appendix A.5.

2.6 Aggregate Business Reallocation and Innovation

The process of business reallocation across firms is central to the dynamics of our model. It is driven by both new entrepreneurs' and incumbent firms' activities. The overall rate of business reallocation, denoted by τ , is given by

$$\tau = \sum_{\theta} e_{\theta} + \sum_{\theta} \sum_n [X(n, \theta) + \nu n] \Psi(n, \theta)$$

where e_{θ} denotes the entry rate into type- θ entrepreneurship, capturing the contribution of entrants to business reallocation, while the second term reflects the incumbent firms' innovation efforts as well as exogenous factors that contribute to business reallocation across firms.

While τ measures the overall flux of business activity, including both new firm entry and incumbent expansion (innovative or otherwise), only some of those changes lead to

direct quality improvements in our model. Thus, τ captures the entire process of reallocation across the economy, encompassing both innovating and non-innovating firms. In contrast, we define τ_g as the *aggregate innovation rate*, which captures the portion of these reallocative events that actually raise product quality and thereby contribute to aggregate growth:

$$\tau_g = \sum_{\theta} s_{\theta}^I e_{\theta} + \sum_{\theta} \sum_n X(n, \theta) \Psi(n, \theta) \quad (8)$$

where s_{θ}^I is a parameter representing the share of entry by θ -type entrepreneurs that leads to a quality improvement upon entry. This parameter is calibrated using empirical evidence to reflect the relative importance of entrants for the overall growth of the economy. The second term is total incumbent innovation. Consequently, τ_g governs growth via successful innovations, and τ governs the broader reallocation of resources across the economy. Because only a fraction s_{θ}^I of entrants raise quality, entry is a more diluted source of growth than incumbent innovation, a tension the policy analysis in Section 5 examines.

Finally, the aggregate growth rate of the economy is expressed as¹⁵

$$g = \ln(\gamma) \tau_g,$$

where $\gamma > 1$ is the step of product quality improvements achieved through innovation. We collect the equations defining the Balanced Growth Path (BGP) equilibrium of this economy in Appendix A.6.

Model Takeaways. Our framework unifies the career choices of individuals (whether to become a production worker, R&D worker, or entrepreneur) with firm-level innovation. Talent allocation across occupations, determined by ability, educational opportunities, and family background, governs both the supply of R&D workers and the emergence of transformative entrepreneurs. The two sides interact in equilibrium: the composition of the entrepreneur population determines R&D demand and the equilibrium wage of R&D workers, which in turn affects who chooses an R&D career and who seeks entrepreneurial opportunities. Financial constraints on education distort this allocation from both sides, reducing the number of R&D workers and transformative entrepreneurs. Section 5 studies policies that act on each side of this market in turn, and quantifies how much of each policy's effect runs through the response of the other.

¹⁵The derivation of the growth rate is given in Appendix A.3.

3 Data and Empirical Analysis

For our empirical and quantitative analysis, this paper leverages rich administrative data from Denmark that allows us to observe individual characteristics, career choices, and firm outcomes. After describing our data and key variable definitions, we develop facts on R&D workers, entrepreneurs, and firms. This evidence provides both validation of our theoretical mechanisms and the ingredients underlying the moments in the estimation in Section 4.

3.1 Data Description

The empirical and quantitative analysis in this paper uses detailed micro-level data from Statistics Denmark (DST) for the years 2001–2013. We rely on four datasets that contain (i) individual background and test (IQ) information; (ii) matched employer-employee data with details on wages and occupations; (iii) information on entrepreneur firms and firm-level outcomes; and (iv) patent data to link innovative behavior to individuals and firms.¹⁶

In the empirical analysis, we group individuals into four categories. Our main sample consists of R&D workers, production workers, and entrepreneurs, where entrepreneurs can either be non-transformative or transformative. We categorize as “R&D workers” those individuals who work in occupations with high patenting rates. More precisely, we select all occupations where at least 1% of workers ever file a patent and classify all individuals in those occupations as R&D workers.¹⁷ By identifying R&D workers based on their occupational classifications, we capture a broader set of contributors to innovation, recognizing that not all R&D workers hold patents, yet they still play a vital role in technological development. With this definition, individuals in R&D occupations account for approximately 1.6% of the working population in our data. We categorize production workers as individuals who work but do not have an R&D occupation.

Finally, we define entrepreneurs as individuals identified in the IVPS/IVPE database as primary founders of a firm with at least one employee. The IVPE and IVPS contain, respectively, “people behind personally-owned new companies” and “people behind new companies”. To distinguish transformative entrepreneurs, we aim to capture the intent to innovate through hiring: an entrepreneur is transformative if they hire at least

¹⁶More details on data construction and variable definitions are provided in Appendix B.

¹⁷Different thresholds, e.g., 0.5% or 2%, lead to similar qualitative patterns across R&D workers and transformative entrepreneurs. Appendix B.3 lists the top R&D occupations, which consist of occupations where at least 1% of individuals have a patent.

one R&D worker at any point in the life of the firm.¹⁸ In contrast, a non-transformative entrepreneur is one who does not hire any R&D workers.

3.2 Empirical Analysis

We start by evaluating the occupational choice of individuals against their IQ and backgrounds, and then move to firm dynamics. Two patterns from the introduction anchor the analysis. Figure 1 showed that entrepreneurs are *negatively* selected on IQ, sharply diverging from the positive selection of inventors and R&D workers.¹⁹ Figure 2 showed that this aggregate pattern obscures a crucial heterogeneity: once entrepreneurs are distinguished by their hiring of R&D workers (17% of entrepreneurs), the two types sort oppositely on IQ, with transformative entrepreneurship rising in IQ while non-transformative entrepreneurship falls. This sorting is a central mechanism of our framework, governing how talent is allocated across entrepreneur types and into the innovation pipeline.

Univariate sorting on IQ, however, leaves the underlying determinants unresolved. IQ is correlated with education, and both are correlated with family background, so the gradients in Figures 1 and 2 may reflect any of these channels. The remainder of this section therefore asks which determinants survive once they are assessed jointly, what an entrepreneur’s own prior occupation adds beyond them, and whether the resulting distinction between entrepreneur types shows up in the firms they build.

Given the IQ-based sorting patterns, we next examine empirical specifications that control for additional factors influencing occupational sorting, allowing us to jointly assess the relative importance of various determinants of career choice. We implement a multinomial logistic regression to explore the determinants of career choice for R&D workers, entrepreneurs, and entrepreneur types. We compare the probability that an individual i enters an occupation k against the probability of becoming a production worker,²⁰ as described in equation (9):

¹⁸Because a larger firm has more opportunities to make such a hire, this classification could on its own favor firms that grow. Appendix Figure C2 also assigns type using only hires in the first five years and compares growth from year five onward, after type is already fixed; the results are unchanged.

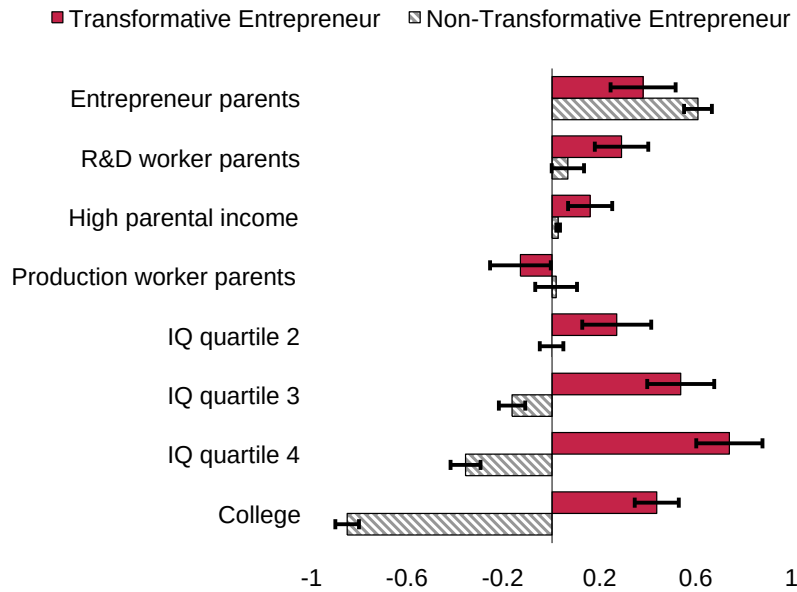
¹⁹This is noted in Akcigit et al. (2025), among others.

²⁰For individuals with multiple occupations during the sample period, we assign each individual i to the occupation k in which they spend the majority of their observed working years. Our results remain consistent when we instead classify occupations based on cross-sectional career observations from a single year. See Appendix B for additional details.

$$\begin{aligned} \log \left(\frac{P_{i,k}}{P_{i,p}} \right) = & \beta_1 \text{college}_i + \sum_{j=2}^4 \delta_j \mathbb{I}\{\text{IQ Quartile}_i = j\} \\ & + \beta_2 \text{entrepreneur parents}_i + \beta_3 \text{R\&D worker parents}_i \\ & + \beta_4 \text{production worker parents}_i + \beta_5 \text{parent income}_i + \Lambda_{b(i)} + \epsilon_i, \end{aligned} \quad (9)$$

where $P_{i,k}$ is the probability that individual i is in occupation k against the probability of being a production worker, $P_{i,p}$.²¹ The independent variables are all binary and indicate individual i 's college attainment, their IQ quartile, whether they have a parent entrepreneur,²² a parent R&D worker, a parent production worker, and whether their parents' income belongs to the top half of the income distribution. We include cohort fixed effects $\Lambda_{b(i)}$, where $b(i)$ is the birth cohort of individual i . We plot the coefficients from a regression with the structure of equation (9) in Figure 4.

FIGURE 4: DETERMINANTS OF ENTREPRENEUR TYPE



Notes: Coefficients from the multinomial logit in equation (9), reporting the relative probability of becoming a transformative entrepreneur or a non-transformative entrepreneur. Regressors are binary indicators for college, IQ quartile, parental occupation, and top-half parental income, with cohort fixed effects. Full estimates in Table C1. Source: DST.

Figure 4 reports the likelihood of becoming a transformative or a non-transformative

²¹Appendix C provides the full details underlying this figure.

²²Parent entrepreneur equals one if either parent is an entrepreneur. This is measured as registered self-employment as a primary occupation, as we do not observe firm founding prior to 2001.

entrepreneur, each relative to the baseline of a production worker.²³ Parental entrepreneurship is a key factor for both types, but IQ and education play a distinctive role for transformative entrepreneurs. In particular, college education and the highest IQ quartile are associated with 55% and 110% higher odds, respectively, that an individual is a transformative entrepreneur, conditional on covariates. These results indicate that transformative entrepreneurs have higher IQs and obtain higher levels of education. In contrast, the probability of becoming a non-transformative entrepreneur is negatively associated with IQ. This result is also consistent with the model, which suggests negative selection by talent into non-transformative entrepreneurship. Parental income is not a strong predictor of entrepreneurship overall conditional on the other covariates, although it is positively associated with transformative entrepreneurship (Table C1), and it may have an indirect impact through the likelihood of college attendance (see Appendix C).

These patterns point to education as an important margin shaping the allocation of talent across occupations. Education raises the odds of becoming both an R&D worker and a transformative entrepreneur. In our model, access to education depends not only on ability but also on family income, because its cost must be paid out of family resources.²⁴ Family background therefore shapes both sides of the market for innovative talent, who becomes an R&D worker and who becomes a transformative entrepreneur, and is a central friction in our analysis.

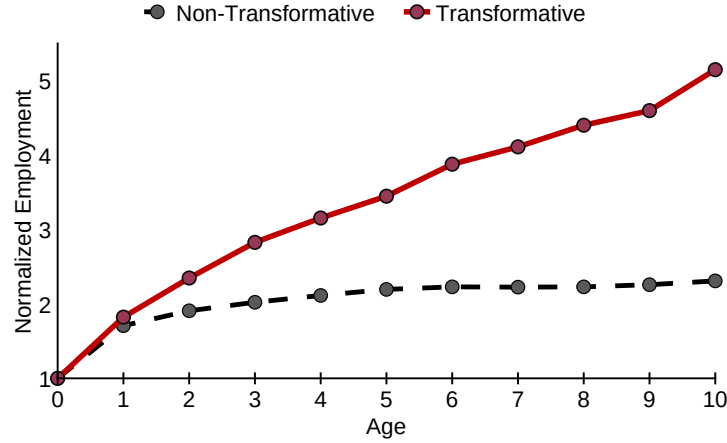
An entrepreneur's own prior occupation is a further strong predictor of their type. Among entrepreneurs who previously worked in R&D, 76% are transformative, compared with only 11% of those who came from production work. Technical training and experience thus appear to equip founders to build and lead innovative firms, reinforcing the role of the R&D career as a pipeline into transformative entrepreneurship. This is the channel through which education operates in our model: schooling matters for transformative entrepreneurship by leading individuals into R&D work.

Firms. Are transformative entrepreneurs more likely to produce a successful firm? Figure 5 documents the evolution of employment of firms held by transformative and non-transformative entrepreneurs over firm ages. To focus on growth rates rather than initial size differences, we normalize employment to 1 at founding for both types, even though transformative entrepreneurs start larger.

²³The corresponding coefficients for R&D workers are reported in Appendix C (Table C1).

²⁴The literature has documented the link between educational attainment and family income in the case of Denmark; see Nielsen et al. (2010) and Akcigit et al. (2025). We document such a link in the context of our data in Appendix C.

FIGURE 5: FIRM EMPLOYMENT BY ENTREPRENEUR TYPE AND AGE



Notes: Average firm employment by firm age for transformative and non-transformative entrepreneurs, normalized to one at founding (age 0). Source: DST.

We find that transformative entrepreneurs start around twice as large as non-transformative entrepreneurs and grow about 9 percentage points faster per year, leading to stark differences in the firm size distribution of the two firm types. As a result, transformative entrepreneurs account for over half of employment by age ten even though they represent a minority of entrepreneurs. This finding underscores their important role in the overall business dynamism in the economy and disproportionate impact relative to their smaller share of initial entrepreneurship. In Appendix C, we show that transformative entrepreneurs, in addition to being more likely to be innovative, also have higher revenue growth and lower exit rates than non-transformative entrepreneurs.

Empirical Takeaways. Overall, our empirical findings suggest that entrepreneurs are negatively selected on ability and schooling, but those who undertake transformative entrepreneurship (by hiring R&D workers) are positively selected, akin to R&D workers, and run larger, faster-growing firms. Further, parental background in occupation and income shapes educational and entrepreneurial opportunities. Taken together, these results underscore the importance of jointly considering individual attributes (such as IQ, education, and family background) and entrepreneurial type to understand firm dynamics and economic growth.

These findings complement the core ingredients of the model: IQ and parental background shape the decision to acquire technical training, while parental entrepreneurship is the strongest driver of entry into entrepreneurship. When an entrepreneurial opportunity arrives, individuals are more likely to become transformative entrepreneurs if they

are educated and have high IQ. Guided by these patterns, we next estimate the model and use it to study counterfactuals and policy.

4 Estimation

In this section, we connect the empirical facts to the model framework and estimate the key parameters of the model governing education decisions, occupational sorting, and firm dynamics. We begin by outlining the estimation procedure, then present the results and evaluate the model’s performance in replicating targeted moments. We also examine how well the estimated model matches untargeted features of the data. The resulting quantification provides the foundation for the counterfactual and policy analysis in Section 5.

4.1 Estimation Strategy and Identification

We begin with a description of the estimation procedure. Our model has 27 parameters: 16 are estimated using Simulated Method of Moments, and 11 are calibrated externally and presented in Table 1.

TABLE 1: CALIBRATED PARAMETERS

Parameter	Description	Value	Source
ρ	Discount rate	0.050	Standard
ψ	Death rate	0.025	40-year career
σ	Curvature on innovation production function	0.500	Akcigit and Kerr (2018)
σ_z	SD of talent distribution	0.224	Data
σ_y	SD of family income distribution	0.619	Data
$\rho_{z,y}$	Correlation of talent and family income	0.144	Data
μ_y	Mean of family income distribution	0.469	Nielsen et al. (2010)
p_f	Probability parent entrepreneur	0.080	Data
c_s	Education cost	0.280	Data
γ	Innovation step size	1.250	De Loecker et al. (2020)
s_{θ}^I	Share of entrants with improvement upon entry	0.25	Data

Notes: These parameters are independently identified in the data or literature.

We set the discount rate ρ to 5 percent. Following the microeconomic literature on innovation, we set the curvature of the innovation production function to $\sigma = 0.5$, indicating a quadratic cost function; this captures diminishing returns to labor inputs

in research.²⁵ The death rate ψ is set to 0.025, implying 40 years of expected working life. The innovation step size ($\gamma = 1.25$) indicates that successful innovations increase productivity by 25%, calibrated from markup data from De Loecker et al. (2020); under the model's limit pricing, the gross markup on every product line equals γ , so measured markups map directly to the step size. The share of innovative entrants s_θ^I is directly measured in the data as the share of entering firms that improve upon the median productivity upon entry in their given industry. The education cost parameter (c_s) measures the income a student forgoes while enrolled in higher education, net of the government stipend received during those years.²⁶

We assume that z and y follow a bivariate lognormal distribution with mean parameters μ_z and μ_y , standard deviations σ_z and σ_y , and correlation $\rho_{z,y}$. The standard deviations of talent (proxied by IQ) and family income are measured directly from data, as is their correlation. The talent mean parameter, μ_z , is normalized, as its level is not separately identified. The family income mean parameter, μ_y , is calibrated to match the share of credit-constrained individuals (30%) reported by Nielsen et al. (2010).²⁷ Finally, the probability of parent entrepreneurship is calibrated to $p_f = 0.08$, representing the baseline rate of entrepreneurship in the previous generation.

The remaining 16 parameters are estimated jointly by matching 40 moments from the data to model-generated moments, which are presented in Table 3 and Figure 6. We combine individual-level moments with firm-level moments to connect these two core features of our framework. These moments focus on ability distributions across occupations, the performance of transformative and non-transformative entrepreneurs, and the aggregate growth rate.

To begin with, we target the shares of R&D workers and entrepreneurs by IQ deciles to discipline the parameters that determine the returns to each occupation, such as α_s . Motivated by our empirical evidence, we let the probability of drawing a type $\bar{\theta}$ and becoming a transformative entrepreneur depend on both ability z and investment in schooling s , according to the following functional form:

$$\Pi(\bar{\theta}|z, s) \equiv \mathbb{P}(\theta = \bar{\theta}|z, s) = \min \left\{ 1, \frac{\varphi + \iota \times s}{1 + \exp(-\kappa(z - \zeta))} \right\},$$

where φ , κ , ζ , and ι are parameters that we estimate by targeting the share of transforma-

²⁵See Akcigit and Kerr (2018) and Acemoglu et al. (2018), who discuss this evidence in more detail.

²⁶We compute c_s as four years of the median production worker's earnings, net of the stipend, which covers roughly a quarter of that income.

²⁷They study the effect of student aid on college enrollment using the 1988 Danish grant reform. To account for borrowing constraints, their analysis classifies as potentially constrained the families whose liquid assets fall below one month's income, about 30% of the sample.

tive entrepreneurs by IQ deciles and by education. φ captures the baseline probability of transformative entrepreneurship, while ι and κ govern the role of education and ability respectively, and ζ represents a threshold parameter. We estimate $\bar{\theta}$ by matching the life-cycle growth profile of transformative entrepreneurs, captured through the average firm size by age. We allow non-transformative firms to grow exogenously at rate ν , which we pin down by comparing the growth trajectories of non-transformative and transformative firms by age.

We assume that schooling and parental entrepreneurship background interact additively in the arrival rate,

$$\lambda_{f,s} = \lambda_f + \lambda_s,$$

and we further normalize $\lambda_s = 0$ for uneducated individuals ($s = 0$) such that we only estimate the additional entrepreneurial opportunity due to education.²⁸ The entry cost c_e is drawn from a Pareto distribution with shape parameter ϖ and a scale parameter fixed at 1.²⁹ We identify these parameters by targeting the share of entrepreneurs with and without entrepreneurial parents and the differences in entrepreneurship rates across education groups.

A core component of our firm dynamics framework is the innovation production function in equation (4). To identify the curvature of the innovation production function with respect to the size of the firm η , we run the following firm-level regression, which connects R&D employment to the total workforce for a firm j at time t :

$$\ln(\text{R\&D Worker}_{jt}) = \beta_0 + \beta_1 \ln(\text{Total Worker}_{jt}) + \varepsilon_{jt}. \quad (10)$$

This relationship describes how firm-level innovation scales with size and helps us identify the returns to scale pattern in innovation: if $\beta_1 = 1$, R&D employment scales proportionally with total employment, corresponding to $\sigma + \eta = 1$ in the model. $\beta_1 < 1$, on the other hand, indicates decreasing returns to scale in innovation. This regression coefficient helps identify the parameter η , which governs how the number of product lines contributes to the firm's innovation capacity, conditional on the calibrated value of σ . As a result, our framework allows for decreasing returns to innovation at the firm level, as long as the R&D workforce does not scale one-to-one with firm size. This is consistent with the economy's need to have a robust pool of entrepreneurs to absorb R&D workers without hitting diminishing returns. We now turn to the estimation results to connect the moments to the underlying parameters.

²⁸Only the sums $\lambda_f + \lambda_s$ enter the model, so the split of their common level between the family and education components is not separately identified.

²⁹The arrival rate λ and the scale parameter of the entry cost cannot be separately identified; hence we set the latter to 1.

4.2 Estimation Results

In this section, we present the internally estimated parameters in Table 2 and the moments that identify them in Table 3 and Figure 6.

TABLE 2: ESTIMATED PARAMETERS

Parameter	Description	Value
$\lambda_{f=1}$	Entrepreneurship arrival rate w/ entrepreneur parent (%)	0.03
$\lambda_{f=0}$	Entrepreneurship arrival rate w/o entrepreneur parent (%)	0.01
$\lambda_{s=1}$	Contribution of education to entrepreneurship arrival rate (%)	0.13
$\bar{\theta}$	Scale for innovation function	0.246
κ	Entrepreneurial type draw - ability slope	1.466
ι	Entrepreneurial type draw - education premium	3.116
ζ	Entrepreneurial type draw - ability threshold	3.472
φ	Entrepreneurial type draw - baseline level	0.281
α_1	Wage function - R&D workers	0.491
α_0	Wage function - Production workers	0.447
m	Entrant cohort mass	0.176
η	Product line share in innovation function	0.330
χ	Utility from education	-3.217
ς	Education preference shock scale	0.377
ω	Pareto shape parameter for entry cost	2.571
ν	Exogenous arrival rate of products (non-transformative firms)	0.119

Notes: All parameters are estimated jointly.

We describe some of the key results from our estimation. For the entrepreneurship arrival rate by parent entrepreneurial exposure, λ_f , we find $\lambda_{f=1} = 0.03\%$ and $\lambda_{f=0} = 0.01\%$, which capture the role of family background: individuals with entrepreneur parents are around three times more likely to have an opportunity to become an entrepreneur. The contribution of education to the entrepreneurial arrival rate ($\lambda_{s=1} = 0.13\%$) also indicates that educated workers are more likely to have entrepreneurial opportunities. Education operates through a second channel as well: the parameter $\iota = 3.116$ raises the probability that the idea an entrepreneur draws is transformative rather than non-transformative. Averaged over the ability distribution, this channel adds about 69 percentage points to that probability, conditional on an entrepreneurial opportunity. The estimated wage parameters for R&D workers ($\alpha_1 = 0.491$) and production workers ($\alpha_0 = 0.447$) capture the earnings premium for R&D workers, which is part of what draws higher-IQ individuals into an R&D career. The value for the product line share in innovation ($\eta = 0.330$) implies decreasing returns to scale in innovation:

doubling both R&D labor and the number of product lines raises the innovation rate less than proportionally ($\sigma + \eta = 0.83$).

TABLE 3: TARGETED MOMENTS

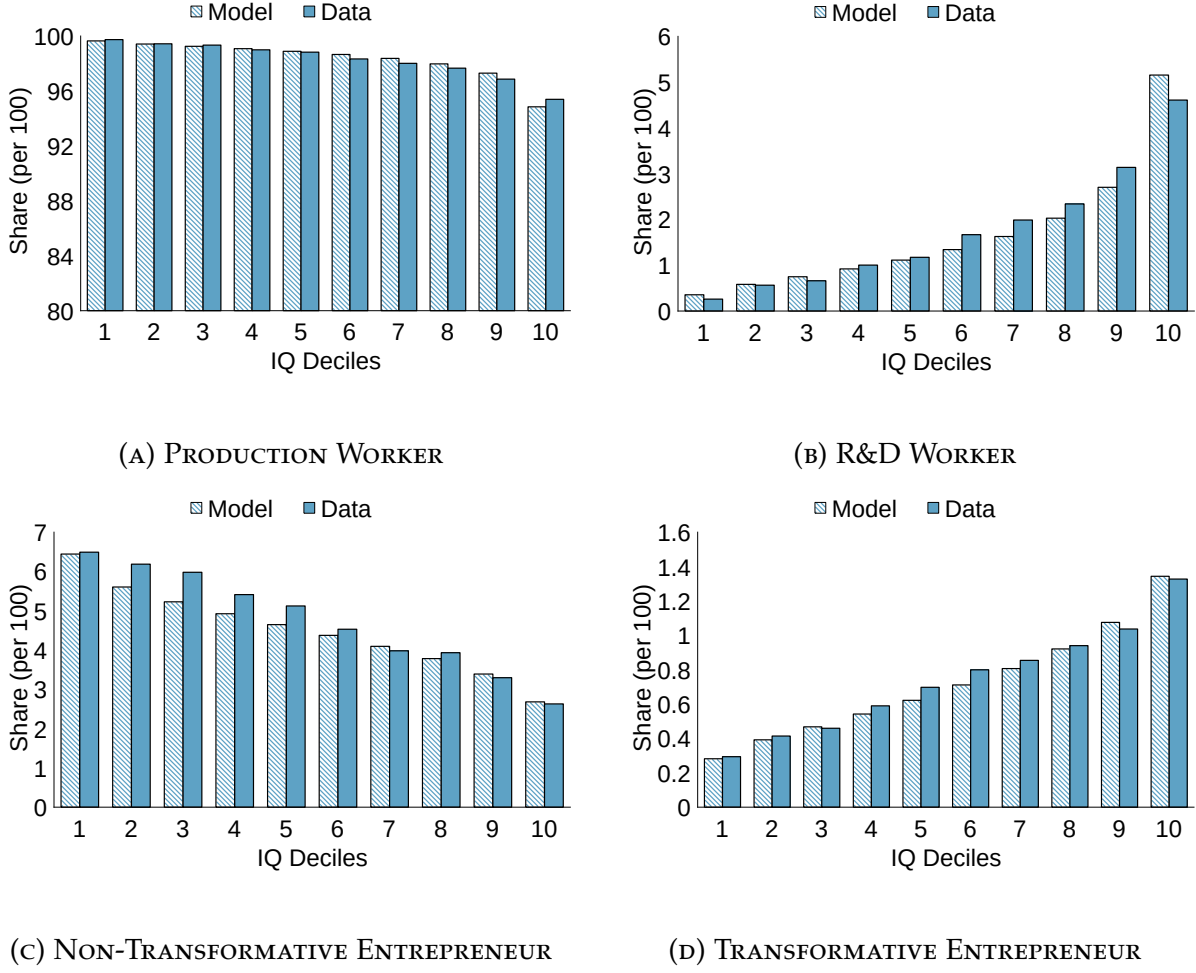
Description	Model	Data
R&D workers share by IQ	In Figure 6	
Entrepreneurs share by IQ	In Figure 6	
Transformative entrepreneur share by IQ	In Figure 6	
R&D workers share in highest family income decile	0.027	0.028
Entrepreneur share by entrepreneur parent	0.10	0.10
Entrepreneur share by non-entrepreneur parent	0.06	0.06
Average firm size at age 10 - Non-transformative entrepreneur	2.23	2.27
Average firm size at age 10 - Transformative entrepreneur	5.03	5.08
Share of transformative entrepreneurs within uneducated individuals	0.006	0.006
Share of transformative entrepreneurs within educated individuals	0.05	0.05
R&D workers log wage premium - highest to lowest IQ decile difference	0.09	0.09
Regression coefficient in equation (10)	0.36	0.38
Aggregate growth rate (%)	1.00	1.00

Notes: Moments jointly input into estimation. Average firm size moments are relative to entry size.

The estimation matches the targeted moments and the occupational distribution. The shares of R&D workers, entrepreneurs, and transformative entrepreneurs by IQ are matched across the ability distribution (discussed below). We match the impact of parental background, with entrepreneurs' children being significantly more likely to become entrepreneurs themselves (0.10 vs 0.06 for non-entrepreneurs' children). The model matches firm growth dynamics: non-transformative entrepreneurs reach 2.23 times their founding size by age 10 (vs 2.27 in the data), while transformative entrepreneurs grow substantially larger (5.03 vs 5.08). The model reproduces the role of education in fostering transformative entrepreneurship: educated individuals, measured in the data as those in R&D occupations, are eight times more likely to become transformative entrepreneurs (0.05 vs 0.006 for the uneducated). We match the change in wage premium over the IQ distribution for R&D workers (0.09 log difference), the elasticity of R&D employment with respect to firm size (0.38 in the data vs 0.36 in the model), and the aggregate growth rate (1%).

On the career choice side, Figure 6 shows the share of individuals by occupation and IQ decile in the estimated model and in the data. Overall, the model provides a close match for the talent allocation pattern across occupations in the data.

FIGURE 6: TALENT ALLOCATION ACROSS OCCUPATIONS



Notes: Share of individuals in each occupation across IQ deciles, in the data and in the estimated model. Source: DST and authors' calculations.

4.3 Untargeted Moments

In this section, we compare our quantified model against untargeted features of the data with respect to occupational sorting and firm dynamics. We study outcomes from the transition of workers to running firms, firm growth, and the effect of education policies using a natural experiment studied in the literature (Nielsen et al., 2010).

First, we examine the transition probability from being a worker to being a transformative entrepreneur, split by worker type, which was not directly targeted in the estimation. The career-choice columns of Table 4 show that, in the data, 76% of the entrepreneurs with a background as R&D workers are transformative, while the remaining 24% are non-transformative entrepreneurs. In contrast, production workers who become entrepreneurs are much more likely to be non-transformative types, while only 11% of

them are transformative entrepreneurs. The table indicates that the quantified model reproduces the much higher propensity of entrepreneurs with an R&D background to be transformative, though it overstates the level for R&D workers (0.90 versus 0.76 in the data).

TABLE 4: UNTARGETED MOMENTS

	— Career Choice —		— Firm Dynamics —			
	Transformative Share		Firm Age 5 size		Firm Age 10 size	
	<i>Data</i>	<i>Model</i>	<i>Data</i>	<i>Model</i>	<i>Data</i>	<i>Model</i>
R&D Worker	0.76	0.90	3.16	3.02	4.35	4.78
Production Worker	0.11	0.11	2.62	1.79	3.04	2.56

Notes: Untargeted moments, data versus estimated model. *Transformative Share* is the fraction of entrepreneurs who are transformative, split by pre-entry occupation. *Firm Age 5* and *Age 10 size* are average employment relative to founding size (age 0). *Source:* DST and authors' calculations.

Next, we examine the dynamic evolution of a firm's size depending on the entrepreneur's occupation background. The firm-dynamics columns of Table 4 report the average firm employment at ages five and ten relative to age 0. The table indicates that, in the data, firms with entrepreneurs who have an R&D worker background grow by a factor of 3.16 after five years and 4.35 after ten years. Firms run by entrepreneurs with a production worker background tend to grow less instead, by a factor of 2.62 after five years and 3.04 after ten. The quantified model broadly matches the patterns along these dimensions.

Finally, we examine the dynamics of firm size by entrepreneurial type. Figure 7 shows the share of firms, in panel (a), and the share of total employment, in panel (b), accounted for by transformative entrepreneurs by firm age.³⁰

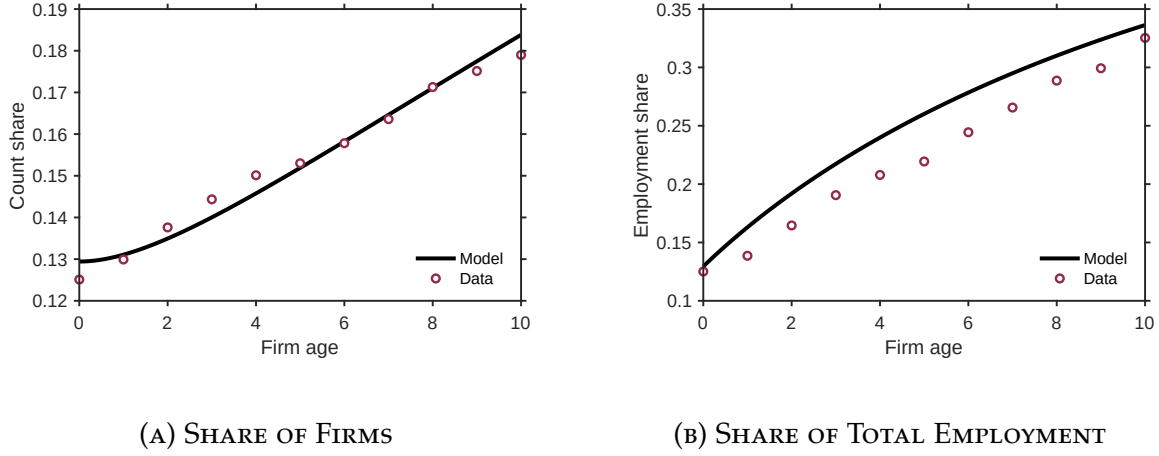
The pattern is monotonically increasing, as transformative entrepreneurs account for a share of firms that increases from about 13% at age 0 to 18% at age 10,³¹ and a share of total employment that increases from about 13% at age 0 to 34% at age 10.³² The age profiles in the model and data are closely aligned, highlighting the increasing importance of transformative entrepreneurs throughout the firm life cycle, which will be central to the policy analysis in Section 5.

³⁰We normalize the initial count of firms to adjust for the share of all individuals in the economy.

³¹Because transformative firms survive longer, their share among entrants (13%) is below the 17% share among all entrepreneurs reported in Section 3, which pools firms of all ages.

³²In this figure, each firm's employment is measured relative to its founding size, as in the estimation, so the employment share at age 0 coincides with the share of firms. Because transformative firms start about twice as large, their share of actual employment is higher: about a quarter at entry and over half by age ten, as reported in Section 3.

FIGURE 7: UNTARGETED MOMENTS: SHARE OF TRANSFORMATIVE ENTREPRENEURS



Notes: Share of firms (panel a) and share of total employment (panel b) accounted for by transformative entrepreneurs, by firm age, in the data and in the estimated model. The initial count of firms is normalized to the share of all individuals in the economy. *Source:* DST and authors' calculations.

The Effect of Education Subsidies. One central component of this paper is the role of education for innovation and entrepreneurship. Section 5.2 studies the effects of education policies through the lens of the quantified model. We evaluate the model's responsiveness against existing empirical evidence in this section. In the main quantitative analysis, we use a single summary statistic from Nielsen et al. (2010) (the 30% share of credit-constrained students); the coefficients those authors estimate from the 1988 Danish stipend reform are not targeted. These can provide a test of whether the model's education choice reproduces the elasticities that they find.

The reform replaced the previous means-tested grant with a universal stipend, raising the level of grants by more than 25 percent for all students above age 19. Nielsen et al. (2010) exploit cross-sectional variation in the size of the stipend change to identify the enrollment response, separately for credit-constrained and unconstrained students. They report average marginal effects from a probit of first-year enrollment on the post-reform stipend S , the pre-reform stipend S^{pre} , a low-assets indicator D , and the interaction $S \times D$, controlling for grade-point average and cohort.³³

We construct the analog on the model's (z, y, f) grid. Pre- and post-reform enrollment probabilities $\Pr(\text{edu} \mid z, y, f)$ are computed in partial equilibrium, holding wages, growth, and firm values at their baseline values, which mirrors the cross-sectional source

³³Their parental income, parental education, sex, age, and geography controls map to our ability index Z in the analog regression. We omit the income percentile (X) and its higher-order interactions ($S \times X \times D$, $X \times D$) because, in the model, X is a deterministic function of the same income state that determines D , generating mechanical collinearity with no counterpart in the Danish data.

of variation Nielsen et al. (2010) use to identify the effect.³⁴

Table 5 reports the comparison. The model reproduces the sign and magnitude of the stipend coefficient (+0.118 in the model versus +0.095 in the data), the negative low-assets coefficient, and the positive interaction stipend $\times$ low assets, which captures the larger enrollment response among constrained students that motivates the use of the reform as an identification strategy. The model’s magnitudes for the interaction effect are larger than their data counterparts, which reflects that the model identifies low assets from income alone, whereas in the Danish data the low-assets indicator is built from parental liquid assets and is only partially correlated with income.³⁵ The model’s overall ability to replicate these patterns without explicitly targeting them suggests that the education side of the model is reasonably well disciplined by the data.

TABLE 5: ENROLLMENT RESPONSE TO THE STIPEND REFORM

	Nielsen et al. (Table 9, Col. 2)	Model (LPM)
Stipend (S)	+0.095 (0.026)	+0.118
Low assets (D)	−0.074 (0.042)	−0.211
$S \times D$	+0.086 (0.038)	+0.238
Ability control (Z)	Yes	Yes
Cohort indicator (R)	Yes	Yes

Notes: The table compares the model against Nielsen et al. (2010): model average marginal effects (weighted linear probability model on simulated data) versus their probit AMEs in Table 9, Column 2. The dependent variable is college enrollment in the data and entry into the R&D track in the model. Model coefficients are rescaled to the probit marginal-effect scale and computed in partial equilibrium; the rescaling, the cell-mass weighting, and the regressors omitted for mechanical collinearity (the income percentile and its higher-order interactions) are detailed in the text. The pre-reform stipend is included but not displayed. Standard errors in parentheses are from Nielsen et al. (2010).

³⁴Because our model outcome (entering the R&D track) is a subset of college enrollment, we report coefficients rescaled by $\phi(\Phi^{-1}(\bar{E}_n))/\phi(\Phi^{-1}(\bar{E}_m))$, the ratio of the standard-normal probit slopes at the two unconditional enrollment rates. This rescaling puts our linear probability coefficients on a scale comparable to the probit specification that Nielsen et al. (2010) estimate.

³⁵This is a feature of our single-resource state y , which carries both the income and wealth roles. We treat the match on sign and order of magnitude as the relevant test; matching the exact level would require introducing a second wealth state, which the rest of our quantification does not require.

4.4 Education and Entrepreneurship

One of the novel components at the center of this framework is the pipeline for transformative entrepreneurs. Before turning to policy, we use the estimated model to quantify how much of this pipeline runs through education. The quantified model suggests education is not just important for the creation of R&D workers, but also for the entry of transformative entrepreneurs. We shut down the effect of higher education on the arrival of transformative entrepreneurial ideas, setting $\lambda_{s=1} = 0$ and $\iota = 0$, which neutralizes the higher-education advantage in the frequency of idea arrival and in the probability of drawing a high-type idea. We then solve for the counterfactual BGP equilibrium under these assumptions.

TABLE 6: IMPORTANCE OF EDUCATION FOR ENTREPRENEURSHIP

Outcome	Baseline	No educational impact	Change
Growth rate (%)	1.00	0.93	−6.62%
<i>Occupation shares (%)</i>			
R&D workers	1.58	1.47	−6.85%
Non-transformative entrepreneurs	1.62	1.66	+2.48%
Transformative entrepreneurs	0.51	0.47	−8.24%

Notes: The counterfactual shuts down the effect of higher education on transformative-idea arrival, setting $\lambda_{s=1} = 0$ and $\iota = 0$. The first two columns report the level in each economy, and the last column the percent change. Production workers, the residual category, are omitted.

Table 6 presents the results of this counterfactual analysis, illustrating the impact of removing the effect of higher education on transformative entrepreneurship. The findings indicate that shutting down this effect leads to a 6.62% reduction in the overall growth rate. This decline can be attributed to several factors. First, the share of transformative entrepreneurs decreases by 8.24%, while the proportion of non-transformative entrepreneurs rises by 2.48%. Second, the share of R&D workers decreases by 6.85%, which combines responses on both sides of the market for innovative talent. The occupational decision of an R&D worker changes because part of the motivation to obtain education and choose an R&D occupation is the expectation that this path increases the likelihood of becoming a transformative entrepreneur. Eliminating the link between higher education and transformative entrepreneurship reduces the perceived benefits of higher education, thereby discouraging individuals from choosing careers in R&D.

In addition, the demand for R&D workers declines, given that there are fewer transformative entrepreneurs. These results emphasize the central role of education in the innovation pipeline, supporting both the supply of R&D workers and transformative entrepreneurship. Appendix D also studies how child background in terms of education and parental occupation shape the life cycle of entrepreneurs' firms.

5 Innovation Policy with Endogenous Supply and Demand for Talent

In this section, we use the estimated model to study a range of innovation policies. Our framework emphasizes that these policies operate on both the demand and supply sides of the market for innovative talent. A large literature studies demand-side instruments, most notably R&D subsidies, and finds that their effectiveness is limited when the supply of research talent is inelastic (Goolsbee, 1998; Acemoglu et al., 2018). A smaller literature studies supply-side interventions that expand the pool of potential inventors, such as investments in education (e.g., Akcigit et al., 2025). The two sides, however, are typically analyzed in isolation: the demand-side literature takes the supply of innovative talent as given, while the supply-side literature abstracts from the heterogeneous entrepreneurs and firms that employ that talent. Our framework allows us to study the two sides together.

The feedback between the demand side (transformative entrepreneurs) and the supply side (R&D workers) runs through two links. The first is the returns to each career, so a policy affecting one side of the market changes the incentives facing the other. Policies that stimulate firms' R&D demand raise the returns to acquiring innovation-related skills and thereby induce more individuals to acquire them and become R&D workers. The second link is the life cycle: R&D workers are the individuals most likely to become transformative entrepreneurs, so a policy that stimulates the supply side of the market today also builds its demand side tomorrow. Education policy thus does more than expand the supply of R&D workers. By altering educational and occupational choices, it also increases the entry of transformative entrepreneurs, whose firms create additional demand for innovative talent. A policy aimed at one side of the market therefore propagates through the endogenous response of the other.

The same feedback means that each policy depends on the side of the market it does not target. Subsidies to firms' R&D do little when the supply of innovative talent cannot expand, while the gains from education policy are muted when the entry of transformative firms cannot respond. Quantitatively, a substantial share of the aggregate effect of each policy arises through the response of the opposite side of the market. Each instrument, in turn, moves both sides: scholarships expand the supply of R&D workers

and the entry of transformative entrepreneurs, while R&D and entry subsidies raise the demand for innovative talent and strengthen the incentive to acquire the technical skills that R&D work requires. Treating supply- and demand-side interventions as separable therefore misses a central amplification mechanism and understates what each intervention achieves.

In what follows, for each policy we compare the balanced growth path with the policy in place to the balanced growth path without it, at a common fiscal cost across policies: demand-side policies that subsidize firms' R&D or entry (Section 5.1), and supply-side policies that subsidize education (Section 5.2).³⁶ For demand-side policies, we find that targeting transformative entrepreneurs is most effective. For supply-side policies, we find that targeting low-income high-IQ individuals is the best policy. For each policy we report the total effect and the effect when one side of the market is held fixed: education for demand-side policies, firm entry for supply-side policies. The gap between the two measures the synergy between the demand and supply sides of the market.

5.1 Demand-Side Policies

We consider three demand-side policies, each at a common fiscal cost of 0.025% of GDP, financed by a lump-sum tax: a subsidy to incumbent R&D expenses (*R&D Subsidy*), an entry subsidy extended to all entrants (*Uniform Entry Subsidy*), and an entry subsidy restricted to transformative entrants (*Transformative Entry Subsidy*).³⁷ The first raises the demand for innovative talent at existing firms; the other two raise it through the entry of new firms. Table 7 reports the change in the growth rate and in occupation shares under each policy, relative to the baseline.

Targeting transformative entrepreneurs at entry is the most effective policy. The transformative entry subsidy raises the growth rate by 5.06%, twice the gain from the R&D subsidy (+2.39%). The uniform entry subsidy, at +0.10%, generates virtually no growth response despite the same budget. The occupation rows show the underlying reallocation of human capital. The transformative entry subsidy raises the transformative-entrepreneur share by 7.7%, boosts the share of R&D workers by 4.7%, and lowers the share of non-transformative entrepreneurs by 2%. Both the R&D and transformative

³⁶These comparisons leave out the transition from the pre-policy balanced growth path to the new one, and the policies differ in timing along it: education policies raise growth only as newly educated cohorts reach the labor market, while entry and R&D subsidies act on current firms. Our results should therefore be read as long-run effects, once the economy has settled on the new path.

³⁷This budget is around one-half of the annual appropriation of Innovation Fund Denmark, which allocates public innovation grants to Danish firms and startups. We keep the budget deliberately small so that each instrument can be evaluated on its own, separating the income effects of its financing from the effects of the policy itself.

TABLE 7: DEMAND-SIDE POLICIES: GROWTH AND OCCUPATION SHARES

	R&D subsidy	Uniform entry	Transformative entry
Growth	+2.39	+0.10	+5.06
<i>Occupation shares</i>			
Production workers	−0.04	−0.02	−0.08
R&D workers	+2.94	−0.06	+4.66
Non-transformative entrepreneurs	−0.91	+0.92	−1.97
Transformative entrepreneurs	+1.64	+0.53	+7.68

Notes: Entries are percent changes relative to baseline; the growth row is the aggregate growth rate and the occupation rows are population shares. Each policy is financed by a lump-sum tax at a common budget of 0.025% of GDP.

entry subsidies beat the uniform one because they channel resources to firms that innovate rather than to the non-transformative entrants a uniform subsidy mostly funds. The R&D worker row shows the supply side responding: the share rises under the R&D subsidy and the transformative entry subsidy, and is flat under the uniform one, because most of the entrants it funds hire no R&D workers.

What separates the R&D subsidy from the transformative entry subsidy is where a marginal dollar does more for growth, that is, for the rate of quality-improving innovation τ_g . Given to an incumbent, the dollar finances an additional R&D worker whose contribution is held down by decreasing returns, as the firm moves along a concave innovation technology. Used to induce entry, it raises the probability that a marginal individual accepts a transformative entrepreneurship opportunity. The payoff is that probability times the growth the new firm generates upon entry and over its lifetime through innovation. Because the firm enters small, it operates where the returns to its R&D are high. Creating that entrepreneur, however, draws an individual largely out of the R&D workforce, so the economy loses an R&D worker even as it gains an innovating owner. In general equilibrium this opportunity cost is more than offset, because a higher return to transformative entrepreneurship raises the value of the R&D track and pulls more individuals into it, expanding the R&D workforce rather than depleting it. The quantified model nets these forces in favor of the entry subsidy, but the ranking reflects this balance, not decreasing returns alone.

Different Margins of the Growth Response. The main findings presented above incorporate all of the general-equilibrium forces at play in response to policies. To understand which forces drive the outcomes, we decompose the full growth response of each policy along two margins. The *entry response* is the part of the growth response lost when firm entry is frozen at its baseline level, isolating the role of firm entry and dynamics. The *education response* is defined analogously by freezing education at its baseline level, isolating the role of education and occupational sorting. Table 8 reports the decomposition, and we focus on two main takeaways.

First, general-equilibrium forces matter for the instruments. Freezing entry of firms lowers the R&D subsidy's response from +2.39% to +1.42%: without the entry of new firms, the policy achieves only 59% of its full effect. When transformative entrepreneurs are subsidized yet education is not able to expand to incorporate new students, the policy only achieves 23% of its full effect. This highlights the interaction of the demand and supply side of the market. This lesson holds for every policy we study: an instrument evaluated with one side of the market held fixed misses a large part of what it achieves.

Second, R&D subsidies are especially reliant on the education response while transformative entry subsidies are less so. With education frozen, the R&D subsidy delivers only +0.03% of growth: the subsidy raises incumbent R&D demand, but without an education-induced expansion in the supply of R&D workers, the additional demand mostly bids up R&D wages rather than R&D employment, a quantitative restatement of the Goolsbee (1998) critique.³⁸ The transformative entry subsidy retains a +1.18% growth effect, small relative to its full effect but far larger than the +0.03% the R&D subsidy retains. In environments where education expansion is not feasible, the transformative entry subsidy therefore becomes relatively more attractive.

Firm-Level Outcomes. The policies studied in this section have different effects over the firm life cycle. Figure 8 reports, for each demand-side instrument and by firm-age group, the percent change from baseline in the average employment of transformative firms (panel a) and in the share of transformative firms in the cohort (panel b). Bars group firms into young (0–5), middle-aged (5–10), and mature (10–20) years; the baseline is the zero line.

Panel (a) shows how these policies reshape the life cycle of transformative firms. Entry subsidies (transformative and uniform) lower the average employment of transformative firms within each age group, more so for transformative entry and among older firms, as they pull in many young, small transformative entrants and tilt the pop-

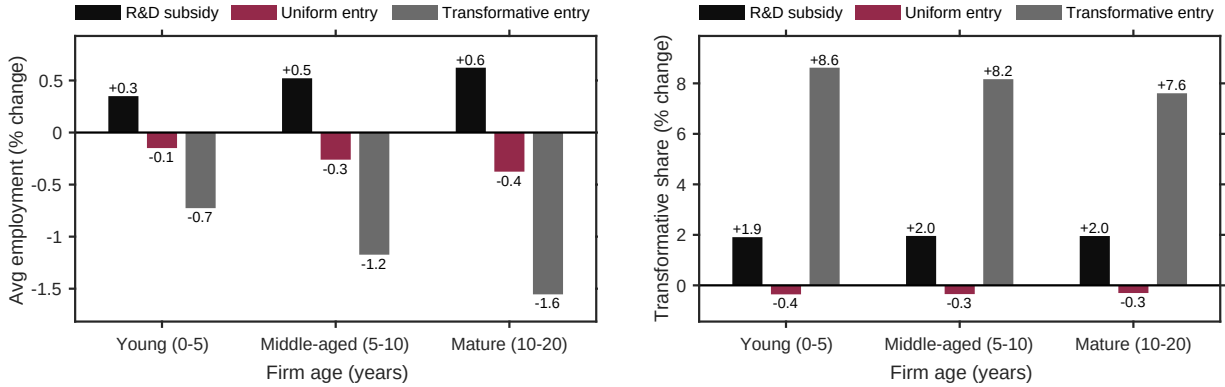
³⁸The education response to the R&D subsidy is also the channel through which Akcigit et al. (2025) operate, in a framework that takes the set of innovating firms as given.

TABLE 8: DEMAND-SIDE POLICIES AND GENERAL EQUILIBRIUM FORCES

Policy	Entry margin			Education margin	
	Full GE	Entry fixed	Entry response	Edu. fixed	Edu. response
R&D subsidy	+2.39	+1.42	+0.97	+0.03	+2.36
Uniform entry	+0.10	-0.01	+0.11	+0.12	-0.03
Transformative entry	+5.06	+0.11	+4.95	+1.18	+3.88

Notes: Entries are percent changes in the aggregate growth rate relative to baseline. *Full GE* is the full growth response, and each margin splits it in two. On the *entry margin*, *Entry fixed* holds firm entry at its baseline level, so growth responds only through the talent pool (worker reallocation), and *Entry response* is the residual once entry adjusts (Full GE minus Entry fixed). On the *education margin*, *Edu. fixed* holds the education decision at baseline, so growth responds only through entry and firm dynamics, and *Edu. response* is the residual once education adjusts (Full GE minus Edu. fixed).

FIGURE 8: FIRM OUTCOMES UNDER DEMAND-SIDE POLICIES



(A) AVG EMPLOYMENT OF TRANSFORMATIVE FIRMS

(B) SHARE OF TRANSFORMATIVE FIRMS

Notes: Grouped bars report the percent change from baseline within each firm-age group (Young, 0–5; Middle-aged, 5–10; Mature, 10–20 years), one bar per demand-side policy, each spending 0.025% of GDP. Panel (a): baseline average employment of transformative firms, relative to founding size, is 2.10, 4.12, and 6.59 across the three groups. Panel (b): baseline transformative-firm shares are 13.8%, 16.8%, and 21.4%.

ulation younger. R&D subsidies, by contrast, raise average employment, increasingly with firm age, since existing incumbents get a greater boost than potential entrants. This carries a lesson on how policy interacts with the firm life cycle: if policymakers are satisfied with the entrepreneurial ecosystem, R&D subsidies boost incumbent size; if they want more novelty and entry, transformative entry subsidies tilt the playing field toward

entrants. Panel (b) shows that transformative entry subsidies make transformative firms more likely across the age distribution. If the policy is persistent, the new steady state has more transformative firms of all ages.

Entry Policies under Imperfect Information. The entry-subsidy gains reported above assume that the planner can perfectly identify transformative entrants and direct the subsidy to them. In practice, type is not observed directly: the planner relies on noisy signals to infer the probability that an entrepreneur would hire an R&D worker for innovation. We illustrate the cost of imperfect classification for the transformative entry subsidy by indexing the planner’s classification ability by a *signal quality* q : with probability q the subsidy offer is directed to a transformative entrant, and with probability $1 - q$ it is allocated randomly across entrants. The perfectly targeted transformative entry subsidy studied earlier in this section corresponds to $q = 1$; the uniform entry subsidy corresponds to $q = 0$. Figure 9 traces the growth response (left axis) and the share of subsidy dollars reaching transformative entrants (right axis) as q varies.

The capacity for the policymaker to extract the correct signal of entrepreneurial type is essential. Growth gains fall from +5.06% at $q = 1$ to +0.10% at $q = 0$, and the share of subsidy dollars reaching transformative entrants falls from 100% to 15.3%. The ranking is therefore conditional on the planner’s targeting technology: policies that can claim near-perfect targeting through observable proxies (for instance, sustained R&D intensity) recover most of the gain, while coarse instruments collapse the ranking back to uniform.³⁹

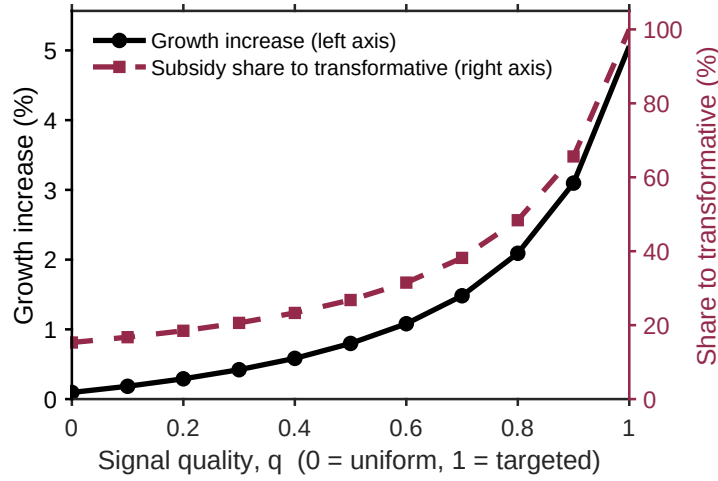
5.2 Supply-Side Policies

The demand-side instruments studied in Section 5.1 operate on entry and innovation decisions taken after individuals have chosen an initial occupation (e.g., production worker or R&D worker). They leave untouched the upstream channels of parental background, education, and income that matter for career choice in both R&D and entrepreneurship.

In this section, we examine policies that act directly on this channel, extending the logic of targeting from the demand side of the market to the supply side. Two instruments are widely used to govern access to higher education: test scores that screen on

³⁹A parallel concern could in principle arise for the incumbent R&D subsidy, as non-transformative entrepreneurs might mimic transformative ones by hiring R&D workers. In our model, however, non-transformative firms gain nothing from hiring R&D workers and the subsidies are too small for them to hire R&D workers as production workers, due to the R&D wage premium. In practice, the R&D subsidy may be harder to direct, reaching incumbents whose marginal R&D is unproductive and where it is absorbed into higher wages rather than innovation. Richer environments could raise further complications that our model does not capture.

FIGURE 9: ENTRY POLICY OUTCOMES BY SIGNAL QUALITY



Notes: Effect of the transformative entry subsidy as the planner's *signal quality* q varies, at a fixed budget of 0.025% of GDP. q indexes the planner's signal quality: with probability q the offer is directed to a transformative entrant, and with probability $1 - q$ it is allocated randomly across entrants. The left axis plots the change in the aggregate growth rate relative to baseline; the right axis plots the share of subsidy dollars reaching transformative entrants. Authors' calculations.

ability and determine entry into university (Edwards et al., 2012), and scholarships that target low-income students (Czarnecki et al., 2021). The quantified model lets us ask how these policies shape not only who becomes an R&D worker but also who becomes a transformative entrepreneur, and through both, economic growth.

We study a menu of targeted full-scholarship programs at the same budget as the demand-side instruments, and then decompose how much of the resulting growth gain comes from the supply side alone and how much from the entry of transformative firms. Appendix D reports a benchmark counterfactual in which the friction is removed altogether.

Targeted Scholarship Policies. We now turn to feasible policies that subsidize education for specific groups. We fix the government's scholarship budget at 0.025% of GDP and use it to fund full-cost scholarships for individuals drawn randomly from a target group until the budget is exhausted. What varies across rows of Table 9 is the targeting rule. *Untargeted* draws beneficiaries from the unconditional population. *High IQ* restricts eligibility to individuals above the median of the ability distribution. *Low income* restricts eligibility to individuals from parental-income backgrounds below the median, the group for which the education friction binds (see Appendix D). *Low income + high*

IQ combines the two screens. Because the budget is fixed, every rule funds the same total number of scholarships; what varies is how many of those recipients are induced to enroll who would not have done so otherwise (the quantity margin) and how able the recipients are on average (the quality margin). Targeting on income concentrates spending on individuals for whom the financial friction binds and so maximizes induced enrollment; targeting on *IQ* concentrates spending on individuals whose education translates most directly into R&D work and transformative entrepreneurship, but a larger share of them would have attended anyway. The first two columns of the table make this tradeoff explicit.

TABLE 9: OUTCOMES FROM TARGETED SUPPLY-SIDE POLICIES

Target group	Avg. <i>IQ</i> educated	Educated share	R&D workers	Transf. entrepreneurs	Δg
Untargeted	−0.78	+9.00	+8.27	+4.84	+6.72
Low income	−1.44	+11.47	+10.28	+6.02	+8.34
High <i>IQ</i>	+0.83	+8.29	+8.49	+4.93	+6.88
Low income + high <i>IQ</i>	+0.74	+10.65	+10.73	+6.23	+8.69

Notes: All entries are percent changes relative to baseline. Scholarship rows fund full-cost scholarships under a common government budget of 0.025% of GDP, with beneficiaries drawn randomly within the target group until the budget is exhausted.

The untargeted policy expands the educated share by 9.00% but lowers the average *IQ* among the educated by 0.78%, because its recipients are drawn from the unconditional population and so are on average less talented than the existing pool of educated workers. Targeting low-income households alone produces the largest quantity expansion (11.47%), because nearly all recipients are constrained at the education margin, but it ignores the quality dimension: average *IQ* falls by 1.44%, since binding income constraints overlap with, but do not select on, ability. The high-*IQ* screen reverses the pattern. It raises average *IQ* by 0.83% but expands the educated share by only 8.29%, because a sizable fraction of high-*IQ* beneficiaries would have attended without the scholarship.⁴⁰ The joint screen captures both margins: it delivers nearly the same average-*IQ* gain as the high-*IQ* screen alone (+0.74%) while expanding the educated share (+10.65%) almost as much as low-income targeting, because the constrained-and-talented subpopulation is precisely the group the friction excludes. This quality–quantity logic was noted for

⁴⁰Equivalently, the share of scholarship-funded attendance that is marginal rather than inframarginal is 79.5% under low-income targeting versus 62.5% under high-*IQ* targeting.

R&D workers in Akcigit et al. (2025); here, we show that the same tradeoff governs the entry of transformative entrepreneurs and the demand for talent as well.

These differences in the number and ability of the newly educated feed through occupational choice and firm entry, shaping the outcomes in the remaining three columns: the R&D worker share, the transformative entrepreneur share, and the growth rate. The low-income and high-IQ screen dominates every other targeting rule on all three margins, delivering the largest growth gain, 8.69%. Two forces explain why. First, low-income individuals are those constrained by financial frictions, so targeting them is the most likely to induce education that would not otherwise occur. Second, high-IQ individuals have the highest return to that education, being the most likely to become transformative entrepreneurs and the most productive R&D workers. The joint screen reaches people who are both constrained and high-ability, so each scholarship both induces education and directs it where the return is highest.

Decomposing the Growth Response. The gains in Table 9 combine two forces: workers reallocate across occupations in response to policy, and firm entry responds in equilibrium to the resulting change in the worker distribution. Table 10 isolates the role of firm entry by holding entry rates fixed at their baseline values and asking how much of each scholarship’s growth gain survives. The first column repeats the full-GE growth gain from Table 9; the second reports the growth gain under fixed entry; the third reports the residual, the *entry response*; the last two columns report the worker and entrepreneur share changes under fixed entry.

TABLE 10: SUPPLY-SIDE POLICIES AND GENERAL EQUILIBRIUM FORCES

Target group	Growth			Shares, entry fixed	
	Full GE	Entry fixed	Entry response	R&D workers	Transf. entrep.
Untargeted	+6.72	+3.79	+2.92	+5.90	+0.81
Low income	+8.34	+4.69	+3.65	+7.59	+0.99
High IQ	+6.88	+3.91	+2.97	+5.17	+0.83
Low income + high IQ	+8.69	+4.90	+3.79	+6.66	+1.04

Notes: Entries are percent changes relative to baseline; *Entry response* is the residual (Full GE minus Entry fixed), in percentage points. *Entry fixed* holds the flow of new entrepreneurs of each type at its baseline level, with the scholarship offer rate held at its full-GE value. Both share columns are measured in the entry-fixed equilibrium: worker shares respond to the policy, while entrepreneur shares are the stock of firms implied by the frozen entry flow.

With entry fixed, somewhat more than half of each policy's growth gain survives; the entry response contributes the remainder, about 44% of the full general-equilibrium effect. The ranking of targeting rules is the same in both columns, and the more growth a rule delivers with entry fixed, the larger the entry response it adds. This reflects the complementarity between the two sides of the market, which runs through two channels. First, a larger supply of R&D workers makes them cheaper to hire, which raises the value of entering as a transformative firm and turns some individuals who would otherwise have stayed employees into founders. Second, R&D workers are the individuals most likely to become transformative entrepreneurs, so a policy that adds to the supply side today also enlarges the pool of potential founders on the demand side tomorrow. The low-income and high-IQ screen is the most effective on both sides of the talent market simultaneously. Supply-side policies thus depend on the demand response in the same way that demand-side policies depend on the supply response: each has its full effect only when the other side of the market can expand.

6 Conclusion

Entrepreneurship is often treated as a single margin of economic dynamism. Our evidence shows why this aggregation can be misleading. In the Danish data, entrepreneurship in the aggregate is negatively selected on cognitive ability. Yet this pattern masks an economically consequential minority: founders of innovation-intensive firms are positively selected on ability and education, hire R&D workers, and build systematically larger, faster-growing, and more innovative firms. Parental entrepreneurship strongly predicts who becomes a founder, whereas ability and education distinguish innovation-intensive founders from other entrepreneurs.

We embed these facts in an endogenous growth model in which occupational choices jointly determine the supply of R&D workers and the demand for them from heterogeneous firms. This organizational link is central: innovative talent contributes to growth not only when individuals acquire the relevant skills, but also when founders create firms able to employ and deploy those skills productively. The policy lesson is therefore not that more business entry necessarily raises growth. In the calibrated economy, support targeted to innovation-intensive entry raises long-run growth roughly twice as much as support for incumbent R&D and substantially more than a uniform entry subsidy. This ranking reflects both the greater innovative potential of these entrants and the endogenous response of the talent market. Policies that expand the supply of R&D workers are more effective when innovation-intensive entry can respond; conversely, policies that stimulate innovative firms are more effective when education can expand the supply

of skilled workers. Indeed, a substantial share of the growth effect of education support operates through the eventual entry of innovation-intensive founders.

The broader implication is that innovation policy must account for both sides of the market for innovative talent. Expanding skills without enabling productive firms to enter and grow leaves part of their potential unrealized; subsidizing firms without an elastic supply of talent similarly limits innovation. Transformative entrepreneurs and R&D workers are thus complementary inputs into the process that turns talent into innovation, firm growth, and aggregate productivity.

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Online Appendix for

Transformative Entrepreneurs: Origins and Impacts on Economic Growth

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A Model Appendix

A.1 Representative Household

Our economy admits a representative household for each cohort and characteristics group: agents of the same cohort with characteristics (z, y, f) pool their income in one household, which insures its members against idiosyncratic career risks such as the arrival of entrepreneurial opportunities, entry cost draws, the evolution of their firms, and death. Each member's career decisions therefore maximize their expected income contribution to the household. This pooling insures risks within a group but does not undo the education friction: schooling must be paid in advance out of parental resources, and a household cannot borrow against its members' future income to finance it. Markets in the model are thus complete over career outcomes but absent at the schooling stage, where family resources alone determine access. The household values the consumption of the final good of its members with the following preferences:

$$U_0 = \int_0^\infty e^{-\rho t} \ln C_t dt, \quad (11)$$

where $\rho > 0$ is the discount rate and C_t denotes consumption at time t . The representative household earns wage income from production and R&D workers, as well as interest income at rate r from asset holdings A_t , which correspond to the total value of firms owned by entrepreneurs within the group. Given this specification, the representative household maximizes utility as defined in equation (11), subject to the flow budget constraint,

$$\dot{A}_t + C_t = rA_t + I_{p,t} + I_{rd,t},$$

where $\dot{A}_t$ is the time derivative of the assets, and $I_{p,t}$ and $I_{rd,t}$ denote the wage income of production workers and R&D workers within the group, respectively. This problem

delivers the standard Euler equation:

$$\frac{\dot{C}_t}{C_t} \equiv g_c = r - \rho, \quad (12)$$

where g_c is the growth rate of consumption, which is equal to the growth rate of the final good, g , in the BGP equilibrium. We next discuss final good production and derive its growth rate.

A.2 Product Market Structure and Static Profits

Consider a representative firm that produces the final good, denoted by Y_t , using a continuum of intermediate inputs $y_{j,t}$. The production technology is Cobb-Douglas in logs:

$$\ln Y_t = \int_{j \in \mathcal{N}} \ln(y_{j,t}) dj.$$

The measure of available product lines is unity, although not all lines are necessarily active at any given time⁴¹. $\mathcal{N}$ denotes the active set of product lines. We normalize the price of the final good to one without loss of generality. The cost minimization problem implies that the representative final-goods producer allocates an identical expenditure, Y_t , to each active intermediate variety j

$$p_{j,t} y_{j,t} = Y_t, \quad \forall j \in \mathcal{N}$$

Intermediate Production. An intermediate goods firm is defined as an entity holding a collection of product lines. These firms, which are run by entrepreneurs in our theoretical framework, are the key objects of interest. Each intermediate good $y_{j,t}$ is produced via a linear technology:

$$y_{j,t} = q_{j,t} l_{j,p,t}, \quad (13)$$

where $l_{j,p,t}$ denotes the effective units of production labor allocated to line j , and $q_{j,t}$ is the productivity level at which the intermediate good is produced by the firm at date t . Consequently, the marginal cost of producing line j is $\frac{w_{0,t}}{q_{j,t}}$, where $w_{0,t}$ represents the wage rate per efficiency unit of production workers.

Firms can augment their portfolios by innovating on a product line. Successful innovation upgrades the quality of intermediate j by a factor $\gamma > 1$, increasing q_j to γq_j . We assume Bertrand competition between the latest innovator and the previous incumbent: upon innovating, the entrant captures the entire market for that intermediate and sets a

⁴¹Product lines may become inactive due to the death of entrepreneurs, for instance.

price equal to the former incumbent's unit cost:

$$p_{j,t} = \frac{\gamma w_{0,t}}{q_{j,t}}. \quad (14)$$

Under this pricing rule, the market is fully served by the new innovator. This implies that each product line j is produced by a single intermediate good firm that has the highest productivity $q_{j,t}$.

Static Profits. The associated static profit for an intermediate line j is

$$\pi_{j,t} = p_{j,t} y_{j,t} - w_{0,t} l_{j,p,t} = \frac{\gamma - 1}{\gamma} Y_t. \quad (15)$$

It follows that the labor demand for line j is

$$l_{j,p,t} = \frac{1}{\omega_{0,t} \gamma},$$

where we define $\omega_{0,t} \equiv \frac{w_{0,t}}{Y_t}$. Note that, at any point in time, this static problem is identical across all product lines a firm owns. If a firm holds n such lines, its total static profit equals $\pi n Y_t$, where $\pi \equiv \frac{\gamma - 1}{\gamma}$; after normalization by Y_t , this is the profit term πn in equation (5). There is no heterogeneity in static profits across entrepreneur types in our framework; differences instead arise in each type's propensity to innovate and expand the firm's portfolio.

Next, we derive the final good output as a function of aggregates. Let $L_{P,t}$ be the aggregate demand for production labor. It satisfies

$$L_{P,t} \equiv \int_{\mathcal{N}} l_{j,p,t} dj = \frac{\phi}{\omega_{0,t} \gamma}$$

where $\phi \equiv |\mathcal{N}|$ defines the measure of active product lines.

As firms set a price equal to $p_{j,t} = \frac{\gamma w_{0,t}}{q_{j,t}}$, we get

$$\begin{aligned}
\ln(Y_t) &= \int_{\mathcal{N}} \ln(y_{j,t}) dj \\
&= \int_{\mathcal{N}} \ln(p_{j,t} y_{j,t}) dj - \int_{\mathcal{N}} \ln(p_{j,t}) dj \\
&= \ln(Y_t) \phi - \ln(w_{0,t} \gamma) \phi + \int_{\mathcal{N}} \ln(q_{j,t}) dj \\
&= \ln(Y_t) \phi + [\ln(L_{P,t}) - \ln(\phi) - \ln(Y_t)] \phi + \int_{\mathcal{N}} \ln(q_{j,t}) dj \\
Y_t &= \phi^{-\phi} L_{P,t}^{\phi} Q_t
\end{aligned}$$

where $Q_t \equiv \exp(\int_{\mathcal{N}} \ln(q_{j,t}) dj)$ is the aggregate productivity index.

A.3 Aggregate Growth Rate

In the previous section, we show that $Y_t = \phi^{-\phi} L_{P,t}^{\phi} Q_t$. Since $L_{P,t}$ and ϕ are constant in stationary equilibrium, the growth rate of aggregate output Y_t equals the growth rate of the aggregate productivity index Q . We can express $\ln Q_t$ after an instant Δt as

$$\begin{aligned}
\ln Q_{t+\Delta t} &= \int_{\mathcal{N}} \left[\frac{\tau_g}{\phi} \Delta t \ln(\gamma q_{j,t}) + \left(1 - \frac{\tau_g}{\phi} \Delta t \right) \ln q_{j,t} \right] dj \\
&= \tau_g \Delta t \ln(\gamma) + \ln Q_t
\end{aligned}$$

where τ_g is the aggregate innovation rate defined in equation (8); innovations are spread uniformly across the ϕ active lines, so each line is improved with probability $\frac{\tau_g}{\phi} \Delta t$, and higher order terms in Δt are omitted. By subtracting $\ln Q_t$ from both sides, dividing by Δt , and taking the limit as $\Delta t \rightarrow 0$, we get

$$g = \frac{\dot{Q}_t}{Q_t} = \lim_{\Delta t \rightarrow 0} \frac{\ln Q_{t+\Delta t} - \ln Q_t}{\Delta t} = \ln(\gamma) \tau_g,$$

where $\dot{Q}_t$ denotes the time derivative of the aggregate productivity index.

A.4 Normalization of Value Functions

In this section, we derive the normalized value function for the firms given in equation (5).⁴² We start with the unnormalized firm value $\tilde{V}_t(n, \theta)$, given by

$$\begin{aligned} r\tilde{V}_t(n, \theta) - \frac{\partial \tilde{V}_t(n, \theta)}{\partial t} = & \pi n Y_t \\ & + \tau n [\tilde{V}_t(n-1, \theta) - \tilde{V}_t(n, \theta)] \\ & + \max_X \left\{ X [\tilde{V}_t(n+1, \theta) - \tilde{V}_t(n, \theta)] - w_{1,t} \left(\frac{X}{\theta n^\eta} \right)^{\frac{1}{\sigma}} \right\} \\ & + \nu n [\tilde{V}_t(n+1, \theta) - \tilde{V}_t(n, \theta)] + \psi [0 - \tilde{V}_t(n, \theta)], \end{aligned} \quad (16)$$

where the term $\frac{\partial \tilde{V}_t(n, \theta)}{\partial t}$ reflects the fact that the value of the firm changes over time. We normalize the value function with the final good Y_t such that

$$\tilde{V}_t(n, \theta) \equiv V(n, \theta) Y_t.$$

Substituting the above definition into the left side of the value function above yields the following,

$$\begin{aligned} r\tilde{V}_t(n, \theta) - \frac{\partial \tilde{V}_t(n, \theta)}{\partial t} &= rV(n, \theta) Y_t - \frac{\partial (V(n, \theta) Y_t)}{\partial t} \\ &= rV(n, \theta) Y_t - \frac{\dot{Y}_t}{Y_t} V(n, \theta) Y_t \\ &= (r - g) V(n, \theta) Y_t \\ &= \rho V(n, \theta) Y_t, \end{aligned}$$

where g is the growth rate of the final good and $r - g = \rho$ from the household maximization problem in equation (12). Finally, by substituting the last term into the unnormalized value function, canceling the final good from both sides, and defining the normalized efficiency wage for R&D workers as $\omega_1 \equiv \frac{w_{1,t}}{Y_t}$, we arrive at the normalized value function in equation (5).

A.5 Firm and Worker Distributions in Equilibrium

This section describes the mass of firms and workers across different states. The stationary equilibrium of the model ensures that the distributions of firms and workers remain

⁴²The normalized value function for workers in equation (3) can be derived similarly and is therefore omitted for brevity.

stable over time.

A.5.1 Firm Size Distribution

The equilibrium mass of firms is denoted by $\Psi(n, \theta, s|z, y, f)$ where firms are indexed by size $n \in \{1, 2, 3, \dots\}$, entrepreneur type $\theta \in \{0, \bar{\theta}\}$, entrepreneur's education s and characteristics (z, y, f) . We denote the relevant marginal distribution, i.e., the version integrated over education and worker characteristics, as $\Psi(n, \theta)$. In stationary equilibrium, the firm mass satisfies a set of conditions that regulate entry, transition, and exit:

$$\begin{aligned} \tau\Psi(1, \theta, s|z, y, f) + \psi \sum_{n=1}^{\infty} \Psi(n, \theta, s|z, y, f) &= e(\theta, s|z, y, f) \\ [X(1, \theta) + \tau + \nu + \psi] \Psi(1, \theta, s|z, y, f) &= 2\tau\Psi(2, \theta, s|z, y, f) + e(\theta, s|z, y, f) \\ [X(n, \theta) + (\tau + \nu)n + \psi] \Psi(n, \theta, s|z, y, f) &= [X(n-1, \theta) + \nu(n-1)] \Psi(n-1, \theta, s|z, y, f) \\ &\quad + \tau(n+1)\Psi(n+1, \theta, s|z, y, f), \quad n \geq 2 \end{aligned} \tag{17}$$

where $e(\theta, s|z, y, f)$ denotes the density of firm entry, given by:

$$e(\theta, s|z, y, f) = \lambda_{f,s} \Pi(\theta|z, s) \times \mathbb{P}[V_e(\theta) - c_e > W_s(z, y, f)] \Phi_s(z, y, f), \tag{18}$$

and integrating it over the entrepreneur's education and worker characteristics gives

$$e_\theta = \sum_{f \in \{0,1\}} \sum_{s \in \{0,1\}} \int_z \int_y e(\theta, s|z, y, f) dz dy.$$

The first equation in (17) describes the entry of firms, ensuring that the inflow of new firms balances the outflow due to business reallocation τ and the mortality of the entrepreneur ψ . The right-hand side represents the entry flow, which depends on the arrival rate of entrepreneurial opportunities and individuals' decision to start a firm. The second equation determines the equilibrium mass of firms of size one. The left-hand side represents the outflow of firms with size one, which can occur either due to firm exit or because they grow into larger firms. The right-hand side captures the inflow into size one, consisting of new firms entering the market and firms that shrink from larger sizes due to business reallocation. The next equation describes a similar evolution of the mass of firms with size $n \geq 2$, where firms experience growth due to innovation or exogenous reasons and shrink due to competitive forces.

Equation (18) defines the density of firm entry with entrepreneur type θ due to individuals with ability z , parental income y , family background f , and education sta-

tus s . The term $\lambda_{f,s}$ captures the arrival rate of entrepreneurial opportunities based on family background and education. The term $\Pi(\theta|z,s)$ denotes the probability of drawing entrepreneur type θ conditional on ability and education. The expression $\mathbb{P}[V_e(\theta) - c_e > W_s(z,y,f)]$ is the probability that the value of starting a firm exceeds the value of remaining a worker. Finally, $\Phi_s(z,y,f)$ is the density of individuals with characteristics (z,y,f) and education status s . Altogether, the equation gives the mass of individuals who receive an opportunity, draw type θ , find it profitable to enter, and have the corresponding characteristics.

A.5.2 Worker Distribution and Occupational Flows

The distribution of workers is represented by $\Phi_s(z,y,f)$, the unnormalized probability density function of worker characteristics (z,y,f) in the production worker or R&D worker occupation, corresponding to $s \in \{0,1\}$. In equilibrium, worker outflows (due to transitions to entrepreneurship and mortality) must be equal to inflows (occupation choice of newborns and transitions out of entrepreneurship). The unnormalized density of workers satisfies:

$$\sum_{\theta} e(\theta, s|z, y, f) + \psi \Phi_s(z, y, f) = m \mathbb{P}(s|z, y, f) \Omega(z, y, f) + \sum_{\theta} \tau \Psi(1, \theta, s|z, y, f). \quad (19)$$

Equation (19) relates the unnormalized density of workers, $\Phi_s(z,y,f)$, in occupations of production or R&D, to two distinct inflows: new entrants making initial occupation choices based on their characteristics, represented by $m \mathbb{P}(s|z,y,f) \Omega(z,y,f)$, and existing entrepreneurs returning to worker status, denoted by $\sum_{\theta} \tau \Psi(1, \theta, s|z,y,f)$. These inflows must exactly offset occupational outflows in BGP, which occur due to the entry into entrepreneurship, captured by $\sum_{\theta} e(\theta, s|z,y,f)$, and worker mortality at rate ψ .

A.6 Equilibrium Definition

In this section, we state the definition of the Balanced Growth Path (BGP) equilibrium from Section 2.

Definition 1 *A Balanced Growth Path (BGP) equilibrium is a collection of value functions $\{W_s(z,y,f), V(n,\theta)\}$, policy functions (e.g., education choice, entrepreneurial entry, R&D intensity), wages $\{\omega_0, \omega_1\}$, and distributions $\{\Phi_s(z,y,f), \Psi(n,\theta)\}$ satisfying the following:*

1. **Workers' Optimization.** *Each agent solves the problem (1) and the Bellman equation in (3), choosing whether to pay the education cost and become an R&D worker or to become a production*

worker, an occupation that does not require education, and whether to transition to entrepreneurship upon an opportunity.

2. **Firms' Optimization.** Each entrepreneur with state (n, θ) solves the firm problem in (5), determining R&D effort (if transformative) to maximize firm value.
3. **Market Clearing.** The supply of production and R&D labor, driven by individuals' occupational choices, matches the total labor demanded by firms, as specified in equations (6) and (7).
4. **Stationary Distributions.** The distributions Φ and Ψ over individuals and firms remain invariant over time, consistent with endogenous transitions and entry/exit, satisfying (17) and (19) in Appendix A.5.
5. **Balanced Growth.** Let $\gamma > 1$ be the quality step per successful innovation and τ_g the aggregate innovation rate. Output grows at

$$g = \ln(\gamma) \tau_g,$$

endogenously determined by equilibrium innovation and entrepreneurship.

B Data Appendix

In the main text, we discussed how to construct data on R&D workers, entrepreneurs, and their firms. This section provides further details on the data construction and occupation definition.

B.1 Dataset Construction and Information

For data on individuals we use the Integrated Database for Labour Market Research (IDA), linked at the person level through the personal identifier (CPR) to Statistics Denmark's income register (IND) for annual income, the education register (UDDA) for highest completed education, and the population register (BEF) for demographics and the parent-child crosswalk. For employer-employee matched data we use the Firm-Integrated Database for Labour Market Research (FIDA); for business origins, the Danish Entrepreneurship Database (Iværksætterdatabasen, IVPS/IVPE); and for patents, data from the European Patent Office (EPO). We connect these to IQ data from Børge Priens Prøve (BPP), the cognitive test administered at the Danish military conscription assessment around age 18. Merging these registers lets us track individuals over their lifetimes across firms and occupations.

B.2 Dataset Details

In this section, we expand on the datasets we combine, the occupational and entrepreneurial definitions, and additional statistics on the main variables of interest. Our work in this project combines various datasets from Statistics Denmark. Here, we discuss each dataset in more detail.

Each individual in Denmark has a unique personal identifier, which lets us link registers at the person-year level. From IDA we take employment status and occupation; annual personal income comes from the income register (IND) and highest completed education from the education register (UDDA), both merged on the personal identifier. Using the population register (BEF) for the parent–child link, we connect individuals to their parents’ income and occupation. We merge in IQ from the BPP for the subset of men who were assessed.

IQ scores are available from 1995 onward, covering most men who entered the workforce or higher education thereafter. We observe BPP scores for roughly 500,000 men, about 30% of the men in our 2001–2013 sample; this share is well below the test’s cohort coverage because our panel also spans older cohorts assessed before the BPP data begin.

The link between individual entrepreneurs and their firms is especially valuable for connecting the individuals behind the firm and their employees. This link is done primarily through the IVPS/IVPE entrepreneurship databases, which provide information on the primary founders of new companies in Denmark. IVPS contains individuals behind companies, while IVPE contains individuals behind proprietorships. We include both as entrepreneurs. Thus, we can observe the progression of individual firms and the individuals behind them. The dataset covers a wide array of firms and individuals. In total, we observe 158,957 unique firms matched to their entrepreneurs from 2001–2013.⁴³

Our entrepreneur sample comprises 147,087 individuals identified as primary founders of a firm with at least one registered employee, of whom 24,281 are transformative. For firm-level analysis we use the entire firm distribution (including entrepreneurs without IQ information); for the IQ×firm analysis we use the 17,669 entrepreneurs for whom we observe both IQ and firm information.

The Firm-Integrated Database for Labour Market Research (FIDA) links each individual to their primary employer (a unique firm identifier) for each year. Firm characteristics are drawn from separate Statistics Denmark registers and merged on the firm identifier: the number of employees, revenues, profits, and industry code come from the Firm Statistics Register (FIRM). Firm age is taken from the firm’s registered founding year. We link individuals and firms to patents from the European Patent Office to mea-

⁴³We define a firm as an entity with a firm identifier and at least one registered employee.

sure innovation; the patent–register match is provided by Statistics Denmark through a name-and-address disambiguation procedure.

B.3 Occupations

As noted in the main text, our analysis focuses on four distinct occupations. In this section, we provide additional details on occupation definitions, particularly for the R&D category, which determines how R&D workers and transformative entrepreneurs are classified.

An R&D worker is defined as an individual who is in an R&D occupation for at least half of their working life. The R&D occupation, in turn, is determined by the patenting intensity of different occupations. Our baseline framework looks at occupations with a patenting intensity of 1% or higher. Table B1 lists the top five most common R&D occupations in the economy, ranked by inventor rate and employment share, respectively. The empirical findings are qualitatively robust to considering alternative cutoffs to classify R&D occupations, such as a patenting intensity of 2% or 0.5%.

Next, a transformative entrepreneur is defined as an entrepreneur who hires at least one R&D worker. Non-transformative entrepreneurs are the remaining set of entrepreneurs who do not hire an R&D worker. In the next section, we show that the firm-level results are robust across variations of this definition, such as hiring an R&D worker within the first five years of operation or ever in the life of the firm.

TABLE B1: R&D OCCUPATIONS

ID	Description	Inventor Rate	Inventor Employment Share
211	Physics and Earth Science Professionals	0.135	0.011
215	Electrotechnology Professionals	0.048	0.007
214	Engineers	0.048	0.188
213	Life Science Professionals	0.041	0.021
231	Professors	0.029	0.083
311	Physics and Engineering Technicians	0.012	0.222
214	Engineers	0.048	0.188
222	Pharmacy and Health	0.011	0.116
200	General Scientist	0.023	0.104
123	Manager (R&D, info, supply)	0.018	0.085

Notes: Top R&D occupations, ranked by inventor rate and by employment share. R&D occupations are those in which at least 1% of workers ever file a patent. *Source:* DST.

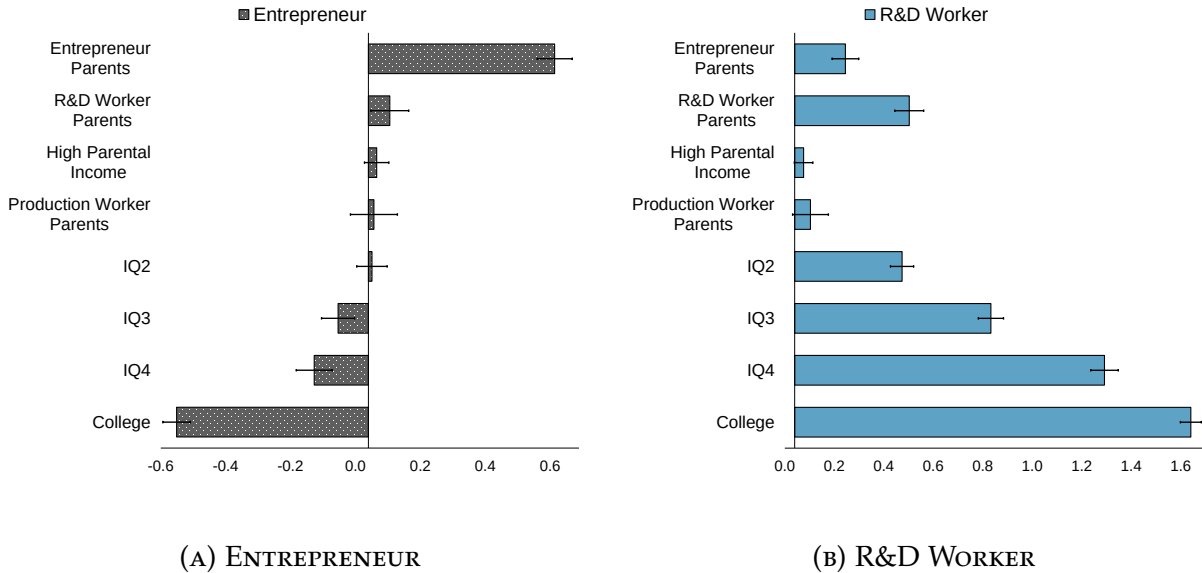
C Empirical Appendix

This section extends the analysis from the main text, focusing on robustness and additional facts on career choice, entrepreneurship, and firm dynamics.

C.1 Career Determinants

This section provides additional results on occupational sorting and transformative entrepreneurship. We start by providing more details on the specification underlying the figures in Section 3. Next, we expand on the role of parental background in education decisions. Finally, we assess the robustness of our empirical findings to alternative definitions of occupations and firms used in the primary specifications.

FIGURE C1: MULTINOMIAL LOGIT FOR CAREER CHOICE



Notes: Coefficients from the multinomial logit in equation (9) for becoming an entrepreneur (panel a) and an R&D worker (panel b), each against the base category of production worker. Full estimates in Table C1. *Source:* DST.

Figure C1a shows that the strongest predictor of entrepreneurship is whether an individual has a parent who is an entrepreneur. The estimated coefficient of 0.574 log odds implies that individuals with a parent entrepreneur have 78% higher odds of being entrepreneurs, conditional on covariates. Those with more education and higher IQ are less likely to be entrepreneurs overall, as the top IQ quartile has 15% lower odds of being an entrepreneur versus a production worker, conditional on covariates.

On the other hand, as seen in Figure C1b, schooling and IQ are both strongly associated with becoming an R&D worker. The top quartile of IQ has 1.25 higher log odds

of being an R&D worker over a production worker, corresponding to 3.5 times the odds. College additionally raises the odds of being an R&D worker rather than a production worker by a factor of five. These results highlight the importance of talent and higher education for becoming an R&D worker, similar to a transformative entrepreneur.

Tables from Main Text Figures. Table C1 displays the coefficients corresponding to the multinomial logistic regressions from the estimation of equation (9) presented in Figures C1 and 4. “Model 1” distinguishes entrepreneurs and R&D workers from production workers. “Model 2” additionally includes the split between transformative and non-transformative entrepreneurs.

Model 1 shows distinct patterns across different occupational choices. For R&D workers, higher IQ is strongly positively associated with choosing this occupation, with coefficients increasing monotonically from quartile 2 (0.434) to quartile 4 (1.252), while parental income on its own has no significant effect. College education has a large positive effect (1.601) on R&D work. By contrast, for entrepreneurs, higher IQ has a negative association, noted in the 3rd (-0.093) and 4th quartiles (-0.167), and college education is strongly negative (-0.591), suggesting average entrepreneurs follow a different human capital accumulation path.

Model 2 separates the two entrepreneurial types. For transformative entrepreneurs, higher IQ is strongly positively associated with choosing this occupation, with coefficients increasing monotonically from quartile 2 (0.270) to quartile 4 (0.740), while parental income shows a significant positive effect (0.159). College education has a substantial positive effect (0.437) on transformative entrepreneurship. Parental background also plays an important role, with parental entrepreneurship (0.380) and parental R&D work (0.291) both having significant positive effects, while having a production worker parent shows no significant association. These results suggest that transformative entrepreneurs represent a distinct group from non-transformative entrepreneurs, with selection patterns more similar to R&D workers regarding human capital requirements but with additional importance placed on parental income, indicating that both ability and access to resources are crucial factors for transformative entrepreneurial activity.

Schooling. Schooling shapes which career an individual can enter. In this section, we focus on two core determinants of schooling: family income and IQ. Both covariates may affect an individual’s propensity to obtain education through ability and financial frictions or opportunities. We perform a simple regression that examines the propensity of an individual to attend college as a function of their family income and IQ. Equation (20) provides the specification, as follows:

TABLE C1: MULTINOMIAL LOGIT ESTIMATES FOR OCCUPATIONAL SORTING

Model 1				
Occupation	IQ Quartile (baseline: Q1)			Parental Income High vs. Low
	Q2	Q3	Q4	
R&D Worker	0.434*** (0.077)	0.793*** (0.074)	1.252*** (0.071)	0.036 (0.039)
Entrepreneur	0.011 (0.029)	-0.093** (0.031)	-0.167*** (0.034)	0.025 (0.023)
Model 2				
	College	Parent Entrepr.	Parent R&D	Production Worker Parent
R&D Worker	1.601*** (0.048)	0.205*** (0.060)	0.463*** (0.043)	-0.064 (0.050)
Entrepreneur	-0.591*** (0.026)	0.574*** (0.033)	0.066 (0.035)	0.017 (0.044)
Model 2				
Occupation	IQ Quartile (baseline: Q1)			Parental Income High vs. Low
	Q2	Q3	Q4	
R&D Worker	0.436*** (0.077)	0.799*** (0.074)	1.261*** (0.071)	0.038 (0.039)
Non-Transformative Entrepreneur	-0.003 (0.030)	-0.167*** (0.033)	-0.361*** (0.038)	0.005 (0.025)
Transformative Entrepreneur	0.270** (0.087)	0.537*** (0.085)	0.740*** (0.084)	0.159** (0.056)
	College	Parent Entrepr.	Parent R&D	Production Worker Parent
R&D Worker	1.606*** (0.048)	0.203*** (0.060)	0.466*** (0.043)	-0.064 (0.050)
Non-Transformative Entrepreneur	-0.855*** (0.030)	0.609*** (0.035)	-0.003 (0.041)	0.084 (0.053)
Transformative Entrepreneur	0.437*** (0.056)	0.380*** (0.083)	0.291*** (0.068)	-0.132 (0.077)

Notes: Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The models are multinomial logit estimations with a production worker as the base category (0). Cohort fixed effects included in both models but not reported. Parent Entrepr. = parent was self-employed, Parent R&D = parent worked in R&D, Production Worker Parent = parent was not R&D or self-employed.

$$s_i = \alpha + \alpha_1 IQ_i + \alpha_2 ParInc_i + \Lambda_{b(i)} + \epsilon_i. \quad (20)$$

Equation (20) is a linear probability model, where s_i is an indicator of individual i 's college attainment (0 or 1); IQ_i is the IQ in percentiles (going from 0 to 1); $ParInc_i$ is parental income in percentiles (going from 0 to 1), based on the income in 2000; $\Lambda_{b(i)}$ is a control for the cohort $b(i)$ of the individual. Table C2 reports the results from the above regression.

TABLE C2: PROBABILITY OF COLLEGE ATTENDANCE

	(1)	(2)
	College	College
IQ	0.389*** (0.002)	0.359*** (0.002)
Parental Income	0.154*** (0.002)	0.121*** (0.002)
Parent College		0.081*** (0.001)
Observations	442587	442587

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Linear probability model of college attendance (equation (20)) on IQ and parental income, both in percentiles (0–1), with cohort fixed effects; column (2) adds a parental-college indicator.
Source: DST.

Table C2 shows that IQ has a substantial positive effect on college attendance, with coefficients of 0.389 and 0.359 in columns (1) and (2), respectively, indicating that moving from the bottom to the top of the IQ distribution raises the probability of college attendance by 38.9 and 35.9 percentage points, respectively. Parental income also raises college attendance, with effects of 0.154 and 0.121. Column (2) additionally controls for parental education, showing that having college-educated parents independently increases a child's probability of attending college by 8.1 percentage points while slightly reducing the magnitude of both IQ and parental income coefficients. These results suggest that, while cognitive ability remains the strongest predictor of college attendance, both financial resources and inter-generational transmission of educational preferences play important roles.

In our model, schooling and R&D occupation coincide by construction, while college attainment is the broader empirical measure. We therefore repeat the regression in equation (20) with becoming an R&D worker as the outcome. Table C3 displays the

TABLE C3: PROBABILITY OF BECOMING AN R&D WORKER

	(1) R&D Worker	(2) R&D Worker
IQ	0.021*** (0.001)	0.021*** (0.001)
Parental Income	0.005*** (0.001)	0.005*** (0.001)
Parent College		0.001 (0.000)
Observations	442587	442587

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Linear probability model of becoming an R&D worker (equation (20)) on IQ and parental income, both in percentiles (0–1), with cohort fixed effects; column (2) adds a parental-college indicator. *Source:* DST.

results for the probability of becoming an R&D worker, with both columns revealing more modest but still highly significant effects. IQ shows a consistent positive effect of 0.021 across both specifications, indicating that higher cognitive ability increases the likelihood of R&D careers, though with a substantially smaller magnitude than its effect on college attendance. Parental income has a small but significant positive effect of 0.005, suggesting that family resources play a role in R&D career selection. In column (2), parental college education shows no statistically significant effect on R&D career choice (coefficient of 0.001), unlike its substantial impact on college attendance. This pattern suggests that while cognitive ability remains important for specialized R&D careers, the inter-generational transmission of preferences and advantages mainly operates through increased college attendance instead of directly influencing occupational selection into R&D work. The overall findings indicate that R&D workers are also primarily selected based on ability and family income.

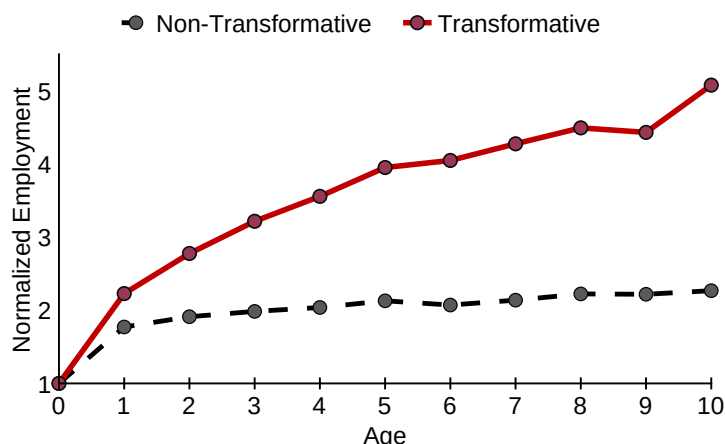
C.2 Firm Dynamics

This section provides more detail on entrepreneurship types. We first compare the life cycles of transformative firms (those hiring R&D workers) and non-transformative firms in Figures C2–C4: transformative firms grow roughly five-fold in employment and seven-fold in revenue over ten years, with somewhat lower exit rates throughout, while non-transformative firms plateau after modest initial growth. We then document how entrepreneurs sort across industries and organizational forms by IQ (Figures C5 and C6).

Figures C2–C4 present a comparative analysis of firm life cycles between transforma-

tive entrepreneurship (defined as firms hiring an R&D worker) and non-transformative entrepreneurship. The employment and revenue figures display measures of growth over the first ten years of operation, with the initial size normalized to one at age zero.

FIGURE C2: FIRM LIFE CYCLE, TRANSFORMATIVE DEFINITION BASED ON FIRST FIVE YEARS



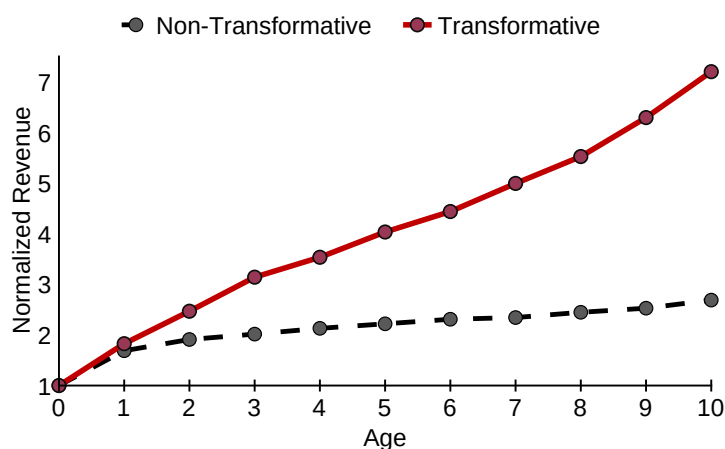
Notes: Average firm employment by age for transformative and non-transformative entrepreneurs, normalized to one at founding (age 0). Here a transformative entrepreneur is one who hires an R&D worker within the first five years (rather than ever). Source: DST.

In Figure C2, we define transformative entrepreneurs as those who hire an R&D worker within the first five years of operation, rather than ever in the life of the firm as in the main text. The two types diverge much as in the baseline: transformative firms grow about five-fold in employment by age 10, while non-transformative firms plateau at roughly twice their initial size after the first year. Even after year 5, non-transformative entrepreneurs barely grow at all ($\approx 1\%$ per year), while transformative entrepreneurs maintain substantial growth ($\approx 5\%$ per year). Because type is assigned within the first five years while the divergence discussed above is measured from year five onward, the comparison confirms that the divergence reflects growth differences by initial type.

Figure C3 shows revenue by firm age. The revenue trajectories closely mirror the employment patterns: by year 10 transformative firms reach more than seven times their initial revenue, while non-transformative firms reach about 2.5 times their starting level. The gap widens after the first year and continues to grow over the decade. Together, the two figures are consistent with transformative entrepreneurs becoming a larger share of the economy with age through creative destruction.

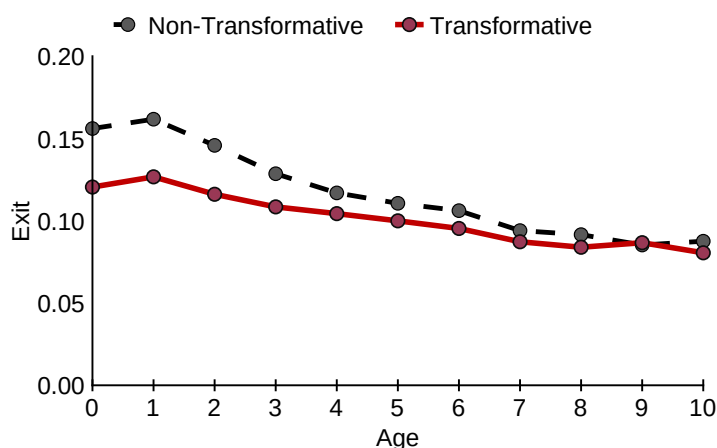
Figure C4 shows exit rates over the first decade, where exit is defined as zero employment or no remaining firm records. Non-transformative firms start with a higher exit rate (about 16% versus 12% for transformative firms), and both types decline with age toward 8–9% by year 8 as surviving firms stabilize. Transformative firms have some-

FIGURE C3: FIRM LIFE CYCLE: REVENUE



Notes: Average firm revenue by age for transformative and non-transformative entrepreneurs, normalized to one at founding (age 0). Source: DST.

FIGURE C4: FIRM LIFE CYCLE: EXIT RATES

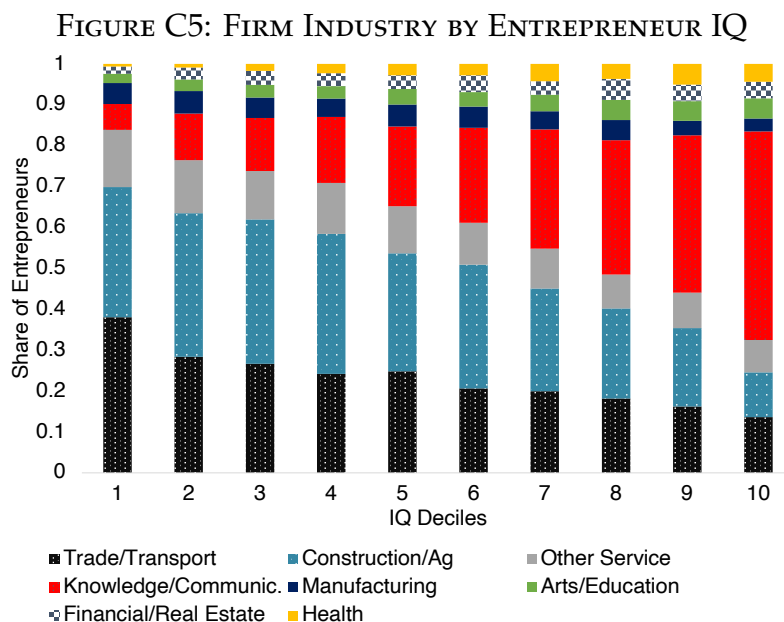


Notes: Firm exit rate by age for transformative and non-transformative entrepreneurs. Exit is defined as zero employment or no remaining firm records. Source: DST.

what lower exit rates throughout, consistent with their R&D investment and growth supporting greater firm stability.

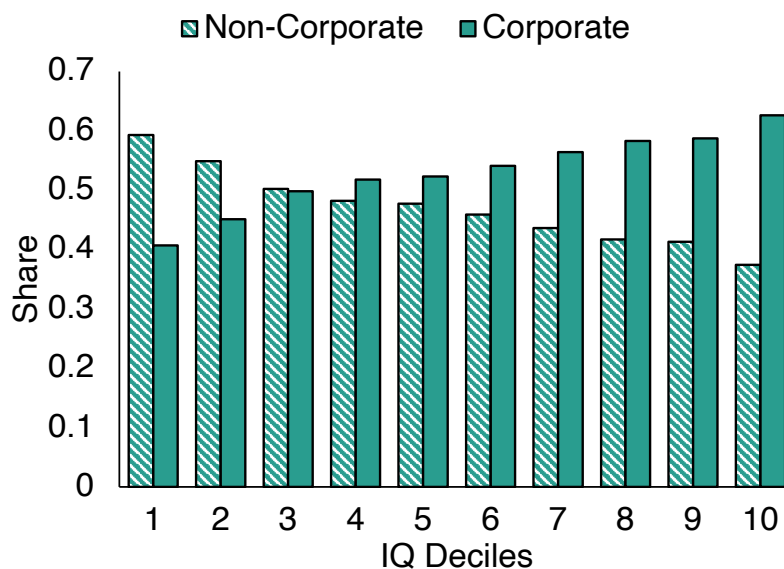
Figure C5 shows how entrepreneurs sort across industries by IQ decile. In the lowest deciles, traditional sectors dominate: Trade/Transport falls from nearly 40% of entrepreneurs in the first decile to about 12% in the tenth, and Construction/Agriculture declines similarly. Knowledge-intensive sectors move the opposite way. Knowledge/Communication rises from a minimal share in the lowest decile to roughly 50% in the top deciles, and Financial/Real Estate, Health, and Arts/Education also increase with IQ. Higher-IQ entrepreneurs thus concentrate in knowledge-intensive industries, consistent with sorting on the returns to cognitive skill. Furthermore, in our data, transforma-

tive entrepreneurs are five times more likely than non-transformative entrepreneurs to file a patent themselves.



Notes: Industry shares of entrepreneurs by IQ decile; shares sum to one within each decile. Source: DST.

FIGURE C6: CORPORATE AND NON-CORPORATE FIRMS BY ENTREPRENEUR IQ



Notes: Share of corporate and non-corporate entrepreneurs by IQ decile; shares sum to one within each decile. Source: DST.

Figure C6 shows the share of corporate and non-corporate entrepreneurs by IQ decile. As IQ rises from decile 1 to decile 10, the share of non-corporate entrepreneurs falls from about 60% to about 37%, with corporate forms taking up the difference. Higher-IQ entrepreneurs are thus more likely to organize as corporations, consistent with our main message: transformative entrepreneurs tend to adopt more formal firm structures geared toward expansion.

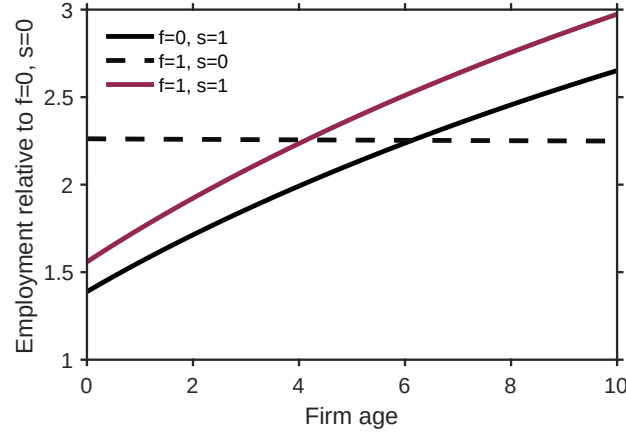
D Quantitative Appendix

This section discusses how education interacts with family background and firm dynamics through two exercises. First, we study the firm life cycle by schooling and parental entrepreneurship background. Second, we focus on what happens to overall growth if we eliminate the frictions in educational attainment.

Firm Life Cycle by Schooling and Parent Entrepreneurship. To further illustrate the role of schooling and family background, Figure D1 presents the impact of education and family exposure to entrepreneurship on the unconditional expected employment creation for an individual by age. The lines represent three groups of individuals: (i) those with parental exposure to entrepreneurship and no higher education ($f = 1, s = 0$, dashed black line); (ii) those with educational attainment but no parent entrepreneurs ($f = 0, s = 1$, solid black line); and (iii) those with education and with parental entrepreneurship exposure ($f = 1, s = 1$, solid red line). Each line plots a group's expected job creation relative to the baseline group with neither schooling nor an entrepreneur parent ($f = 0, s = 0$). Because the expectation is unconditional, averaged over everyone in the group rather than just those who start a firm, it combines two margins: the probability of becoming an entrepreneur and the expected size of the firm they build.

Figure D1 reveals three patterns. First, for individuals without higher education, parental exposure leads to over twice as much job creation throughout the life cycle. This effect is due to the higher propensity to become entrepreneurs for individuals with a parent entrepreneur. Second, we observe that individuals with schooling exhibit a steeper job creation profile. Higher education increases the likelihood that these individuals will draw a transformative type, conditional on having an entrepreneurial opportunity. Educated individuals who draw a low type are therefore more likely to delay starting a firm than uneducated ones, since they expect a better chance of drawing a high type later. As a result, individuals with higher education show lower job creation than those with parental exposure alone in the early years, but overtake them in the central and later stages of their careers. Finally, the fact that individuals with both higher education

FIGURE D1: FIRM LIFE CYCLE BY SCHOOLING AND PARENTAL ENTREPRENEURSHIP



Notes: Unconditional expected employment creation by age, for individuals with a parent entrepreneur only ($f=1, s=0$), education only ($f=0, s=1$), and both ($f=1, s=1$), each relative to individuals with neither ($f=0, s=0$). Each line combines the probability of starting a firm and the expected firm size. Authors' calculations.

and parental exposure to entrepreneurship have the steepest employment creation profile highlights that the interaction of parental entrepreneurship and schooling is central to the contribution of transformative entrepreneurs to job creation.

Eliminating Education Friction. Here we expand on the role of the education friction in occupational allocation and economic growth. We solve for a counterfactual BGP equilibrium in which the financial friction in access to education is fully removed: every individual who wishes to attend school can afford to do so. Individuals still pay the cost of education, which enters their discounted lifetime earnings, but they are no longer financially constrained. Table D1 reports the resulting equilibrium changes: the growth rate rises by 10.96%, the R&D worker share by 14.82%, and the transformative entrepreneur share by 7.89% relative to baseline. The result is consistent with the mechanism quantification in Section 4.4: a substantial fraction of the gap between baseline and unconstrained innovation can be traced to barriers in access to education, which depress both the supply of R&D workers and the entry of transformative entrepreneurs. This counterfactual is not itself a feasible policy, as it shuts down the friction at no cost to the government. Nor is it an upper bound on what supply-side policy can achieve: a well-designed scholarship subsidizes individuals at the education margin and could in principle reallocate talent more efficiently than the frictionless equilibrium on some margins. We report it simply to quantify how much innovation activity is forgone because the friction binds.

TABLE D1: ELIMINATING THE EDUCATION FRICTION

Outcome	Baseline	No friction	Change
Growth rate (%)	1.00	1.11	+10.96%
R&D worker share	1.58	1.82	+14.82%
Transformative entrepreneur share	0.51	0.55	+7.89%

Notes: The counterfactual removes the financial friction in access to education, lifting the requirement that parental income cover the schooling cost ($y \geq c_s$) so that every individual can enroll. The first two columns report the level in each economy, and the last column the percent change.